CREDIT EDA ASSIGNMENT

MODULE 7

STUDENT DETAILS

- NAME – SOUVIK BISWAS
- BATCH – DSC 69
- COURSE- PGP IN DSC & AI

BUSINESS OBJECTIVE

POINTS :

1.This case study aims to give you an idea of applying EDA in a real business scenario. In this case study, apart from applying the techniques that you have learnt in the EDA module, you will also develop a basic understanding of risk analytics in banking and financial services and understand how data is used to minimize the risk of losing money while lending to customers.

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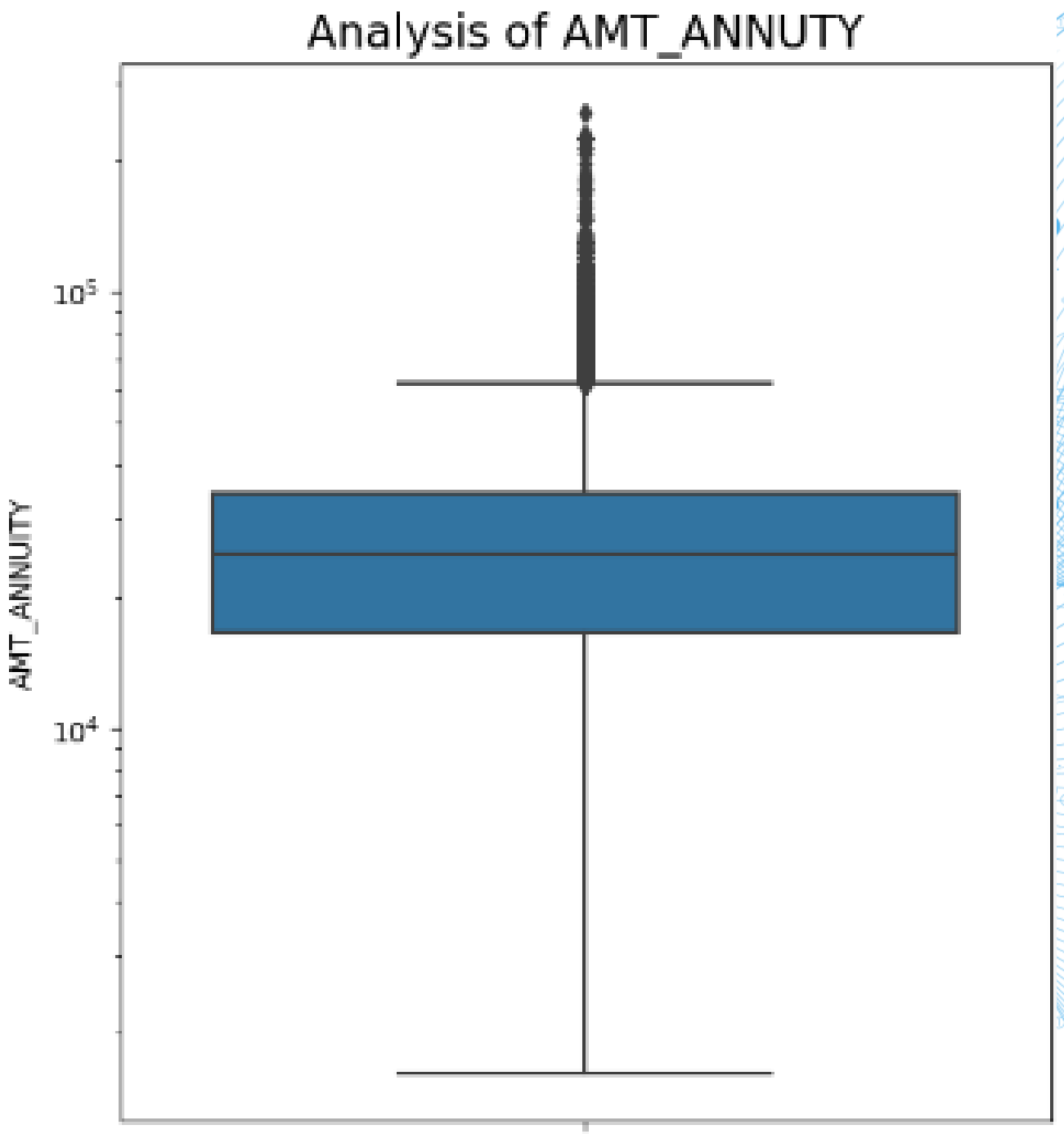
2.In this case study aims to identify patterns which indicate if client has difficulty in paying their installments which may be used for taking actions such as loan denying , reduction of the loan amount , lending at a higher rate of interest. This would ensure that the customer capable of repaying the loan are not rejected. Identification of such customer is the aims of the EDA assignment.

ANALYSIS AND STEPS TAKEN :

- DATA SOURCING
- DATA LOADING AND DATA CLEANING
- UNIVARIATE ANALYSIS
- BIVARIATE AND MULTIVARIATE ANALYSIS

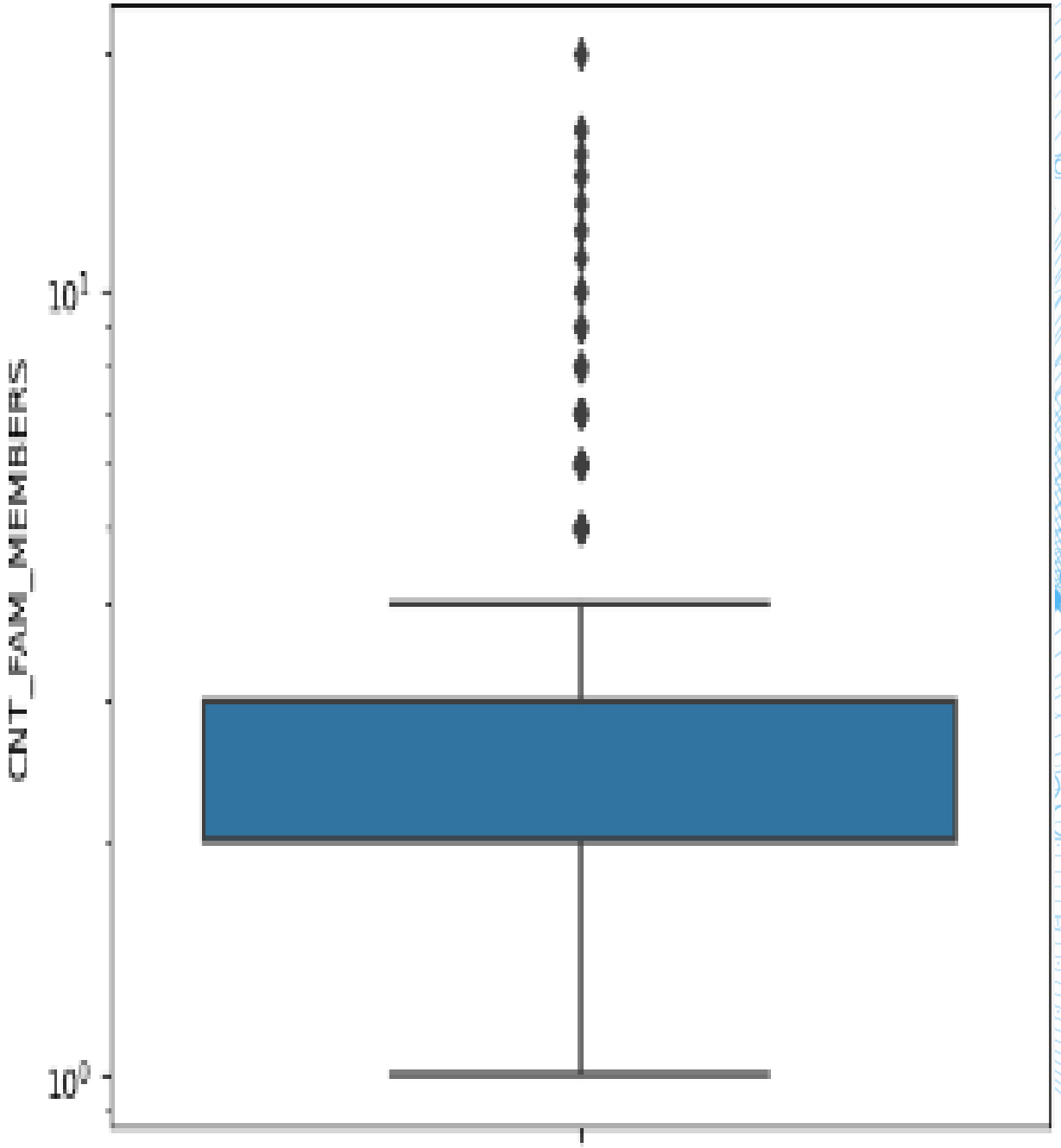
Analysis of AMT_ANNUITY column

From the Box Plot we can see that there are sever outliners and the difference between max and min is quite severe. So we are taking median value to replace those null values.



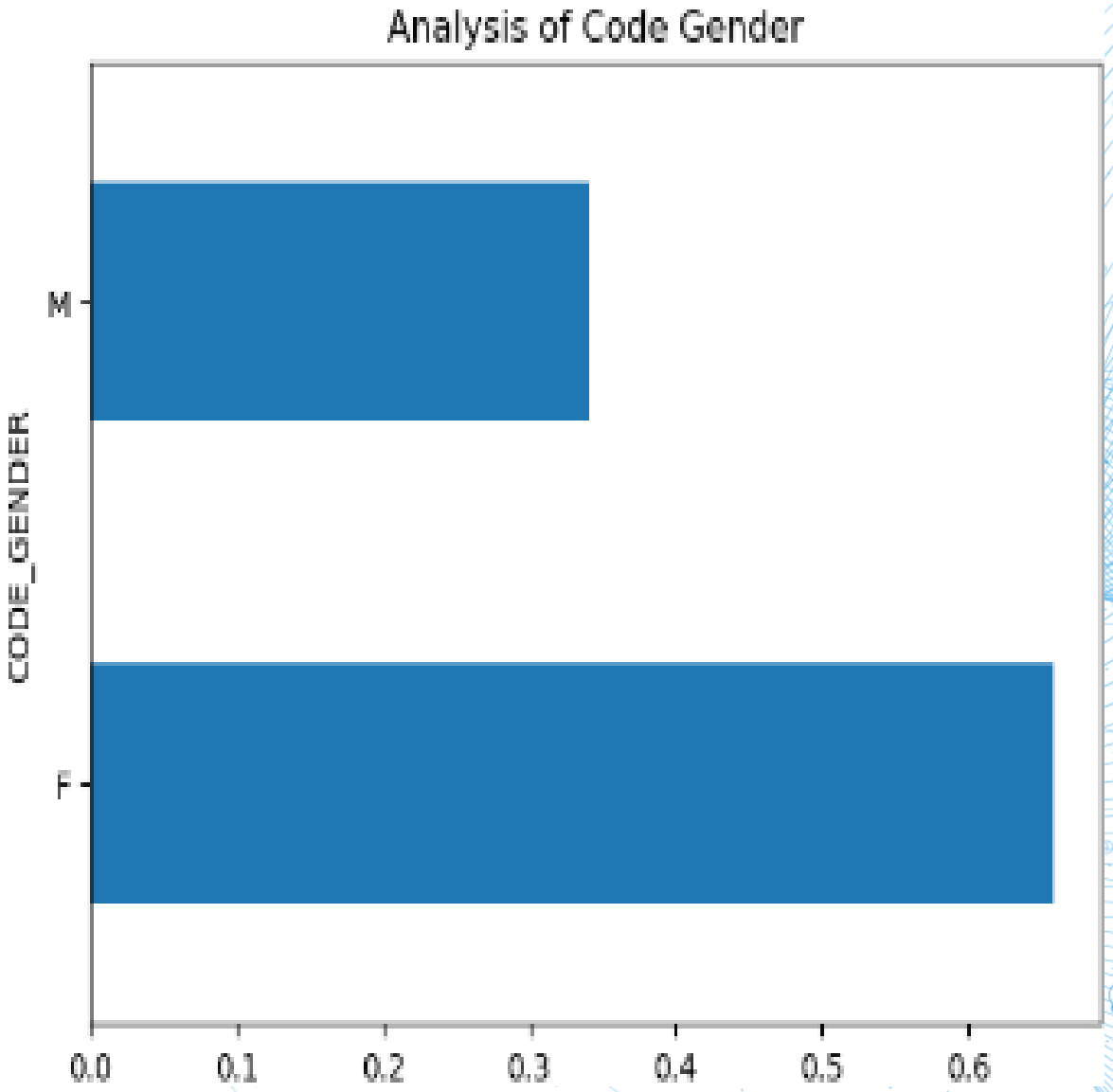
THE VALUE OF COUNT OF FAMILY MEMBERS

From the BoxPlot we can see that there are severe outliners and their is quite a difference between 75% and max.So we are taking median value to replace those null values.



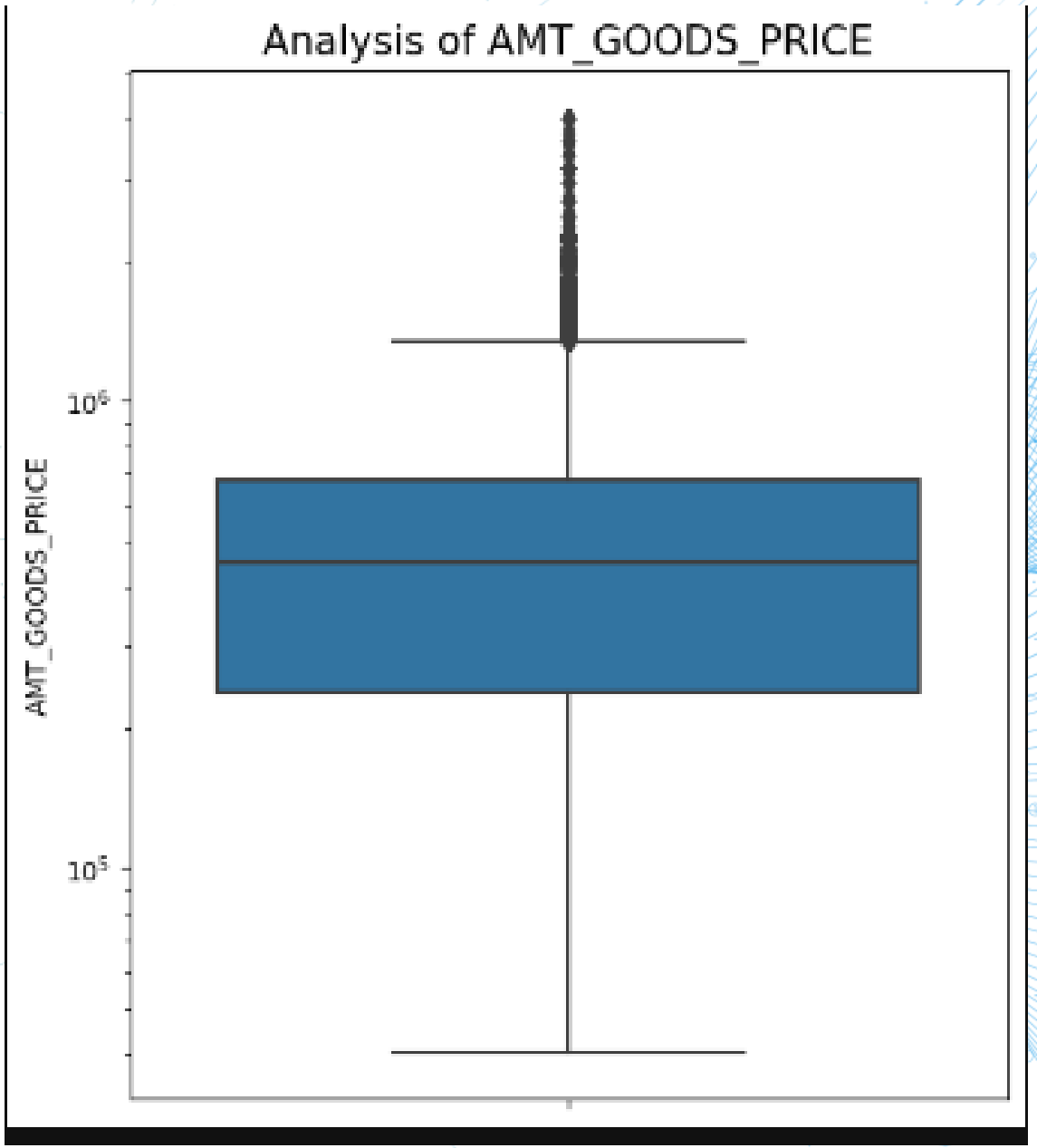
ANALYSIS OF GENDER CODE

As We can see that Female(F) is having the majority and only 4 rows are having XNA values. So, there wont be any major impact in the dataset if we can update those columns with Gender 'F'.



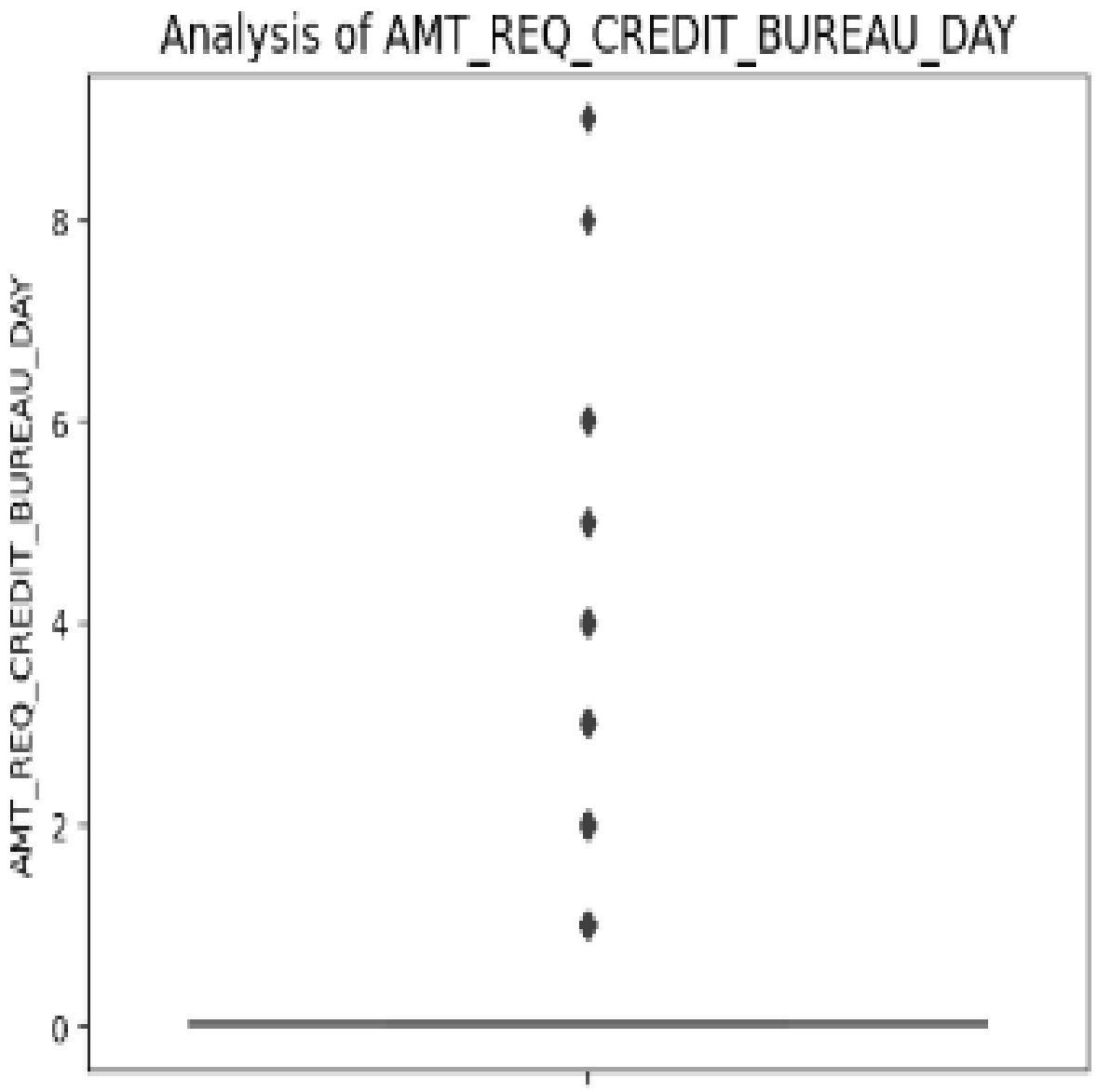
ANALYSIS OF AMT_GOODS PRICE

As we can see we are not getting any clear understanding of the data, so we would keep the null values.



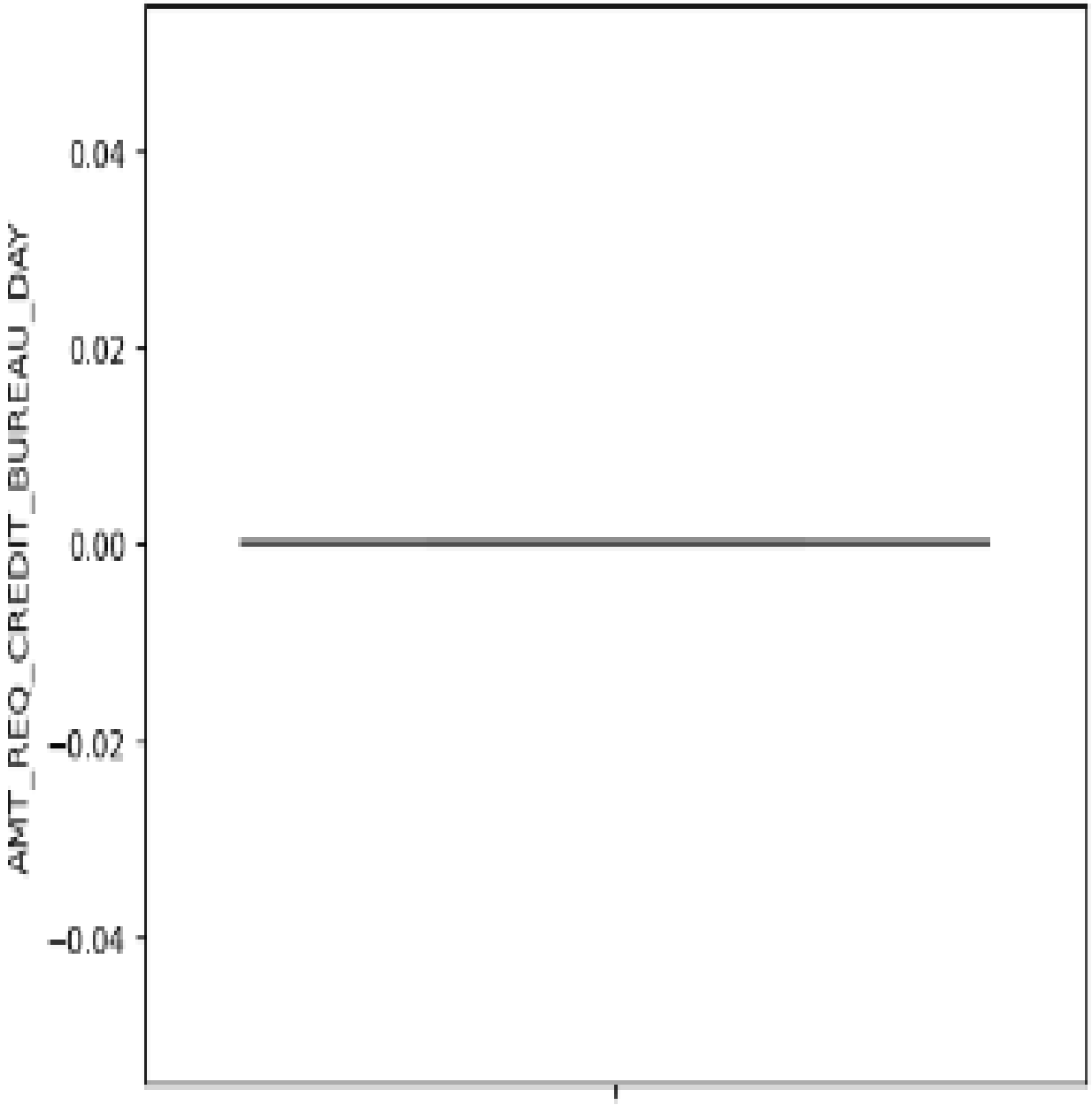
HANDLING OF AMT_REQ_CREDIT BUREAU_DAY

As we can see the numerical column AMT_REQ_CREDIT_BUREAU_DAY has outliers so they need to be removed or capped.Also for missing value fillup we need to use the median in this case.



HANDLING OUTLIERS

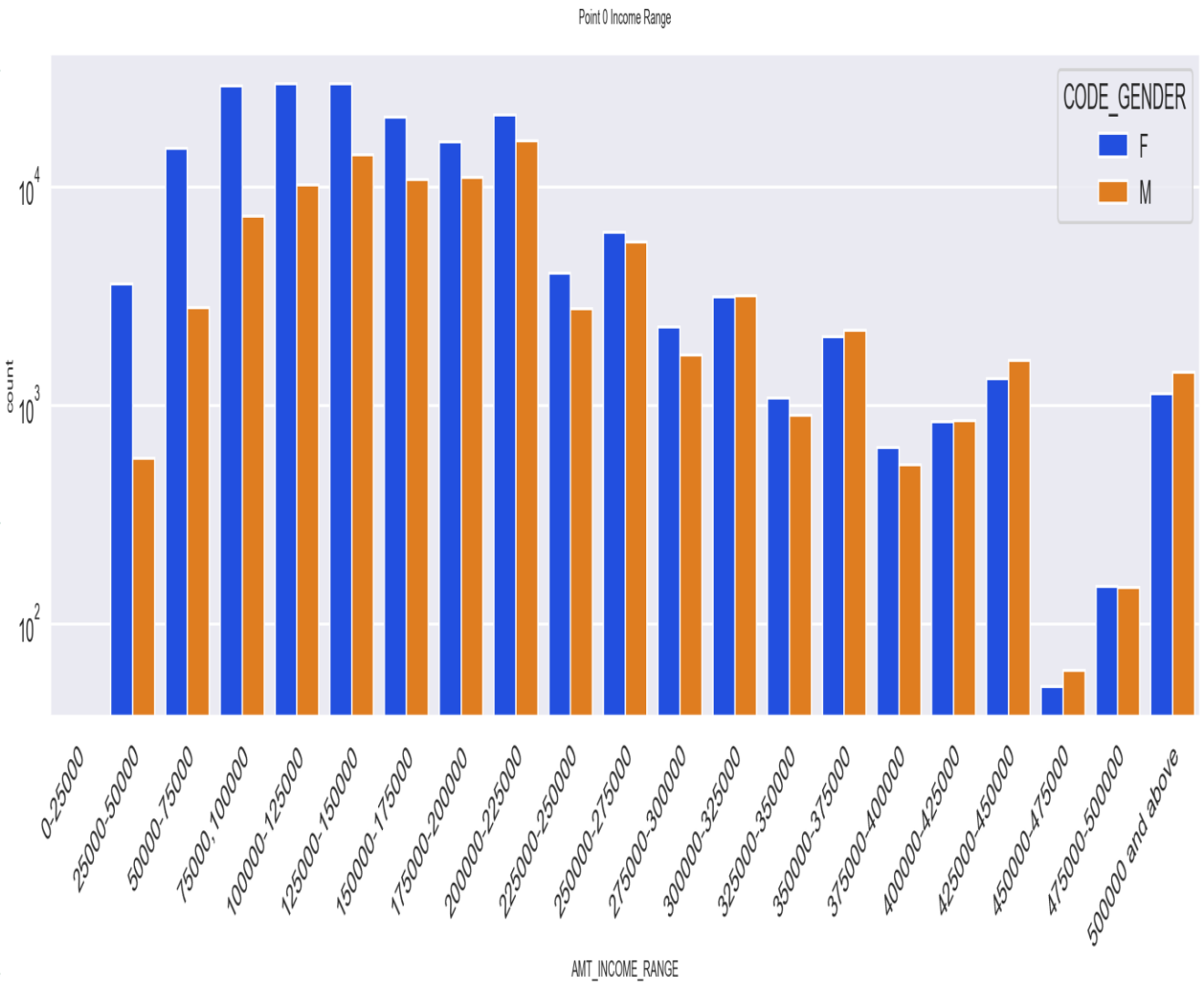
The graph shows a horizontal line at y=0. This indicates that the variable 'AMT_REQ_CREDIT_BUREAU_DAY' has a constant value of 0 across all data points. This suggests that the variable 'AMT_REQ_CREDIT_BUREAU_DAY' is not a useful predictor in a model, as it does not vary and therefore cannot explain any variation in the target variable.



UNIVARIATE ANALYSIS OF CATEGORIES

Points to be concluded from the above graph for Point=0(Non-Defaulters).

1. Female counts are higher than male.
2. This graph show that females are more than male in having credits for that range.

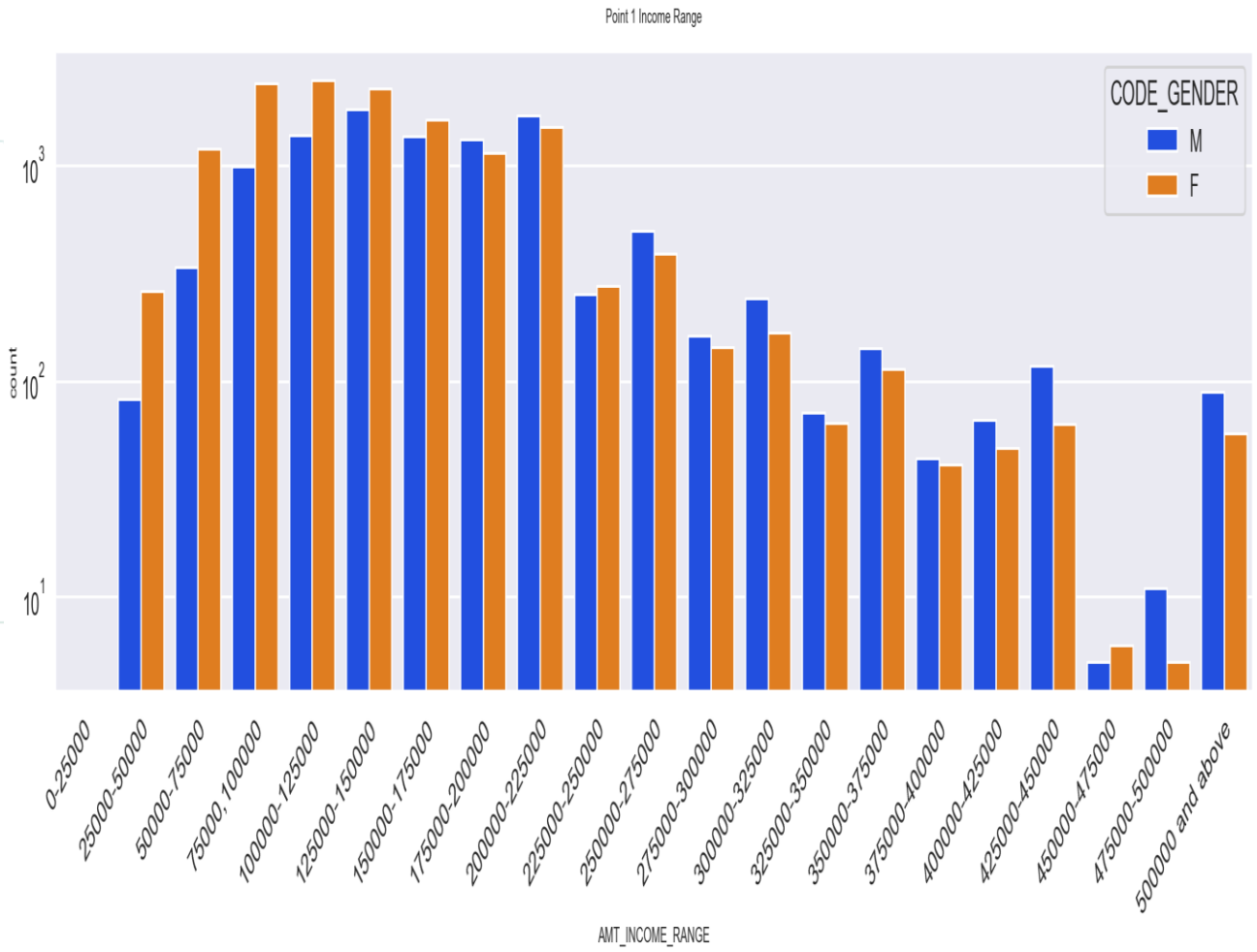


UNIVARIATE ANALYSIS OF CATEGORIES

Points to be concluded from the above graph for Point=1 (Defaulters).

1. Male counts are higher than female.

2. This graph show that males are more than female in having credits between 100000 and 200000.



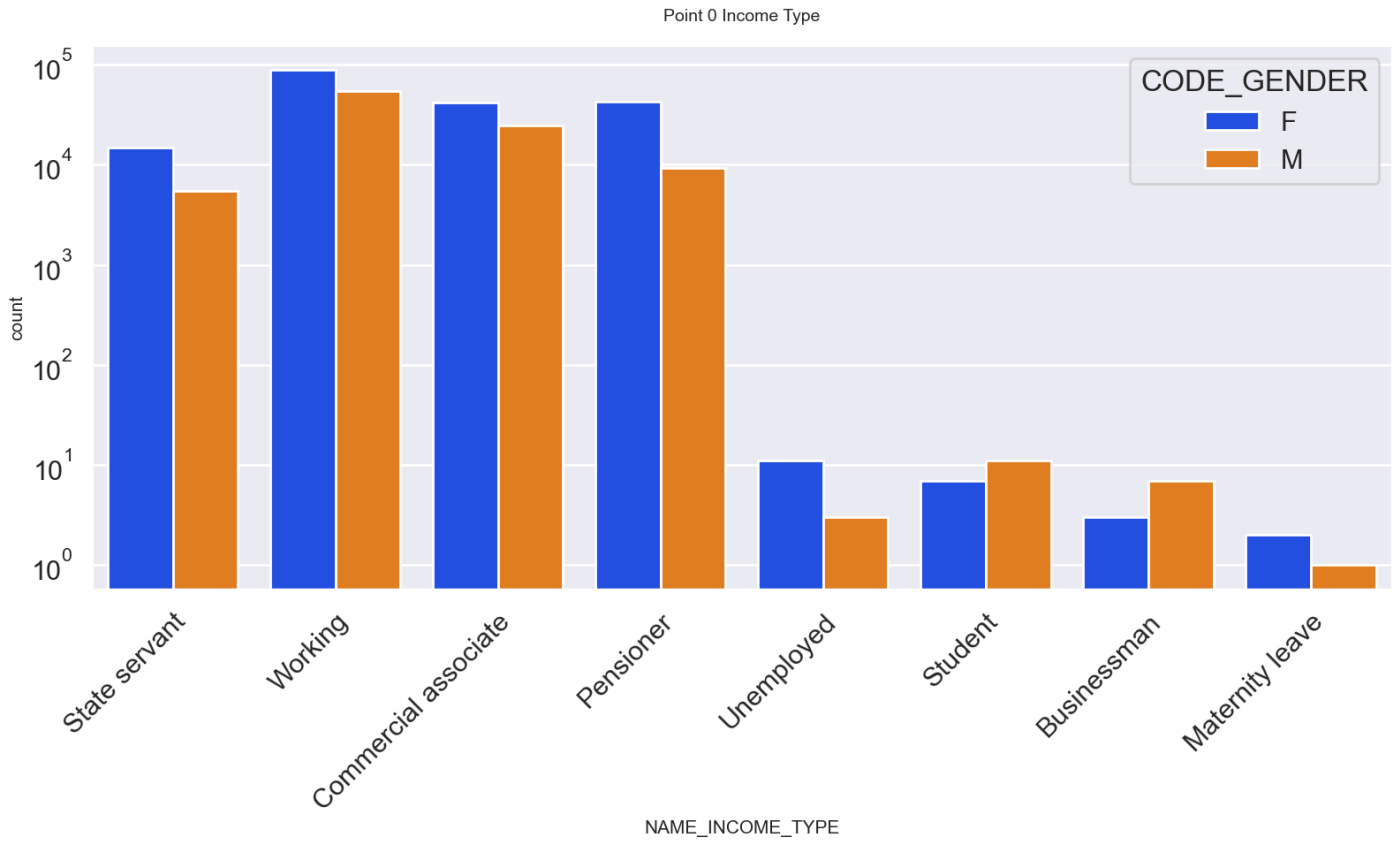
INCOME TYPE FOR POINT 0

Points to be concluded from the above graph for Point=0 (Non-Defaulters).

1. Female are having more credit than males

2. High number of credit for income type working , commercial associate , pensioner and state servant

3. Low number of credit for income type student ,unemployed, businessman and maternity leave

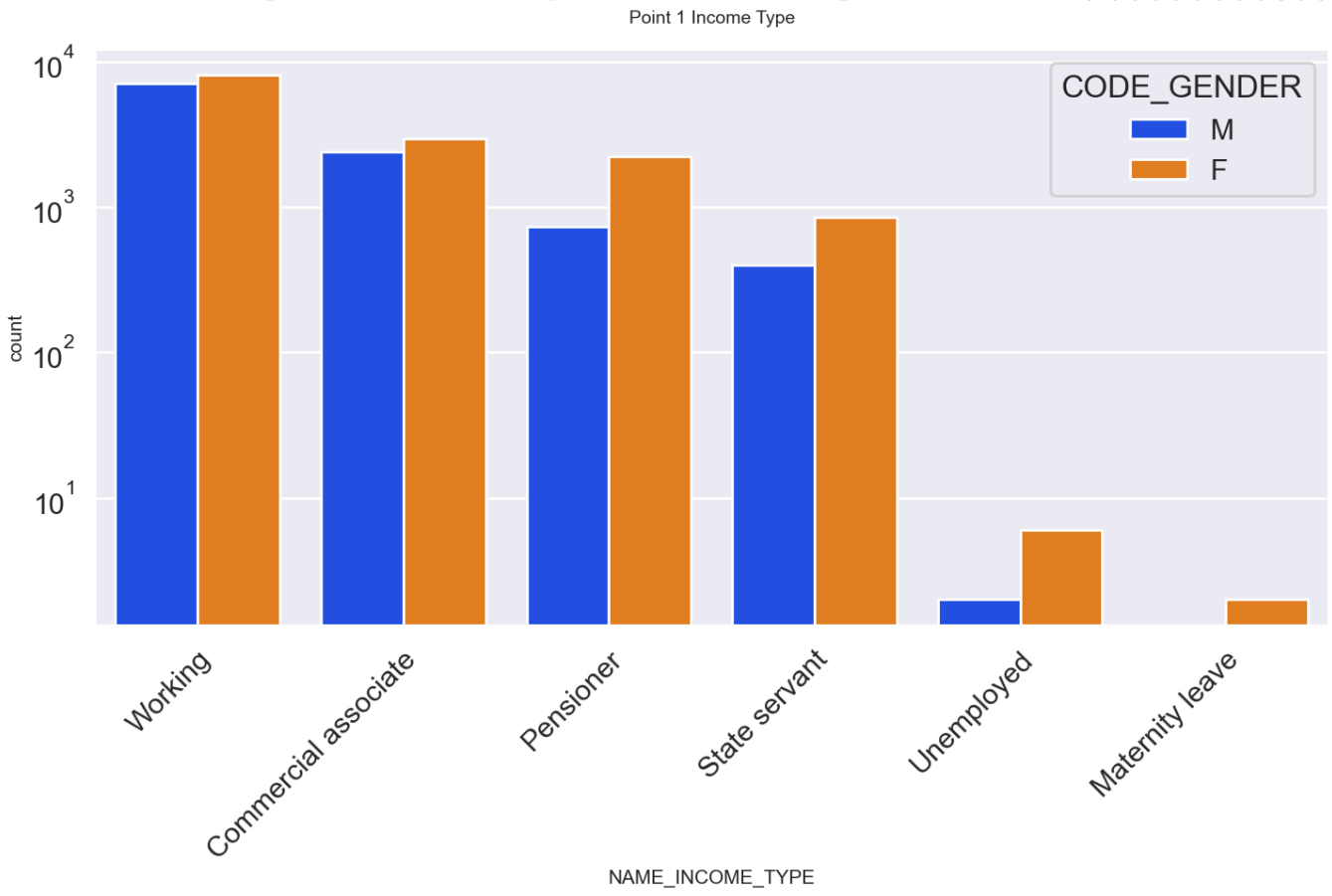


INCOME TYPE FOR POINT 1

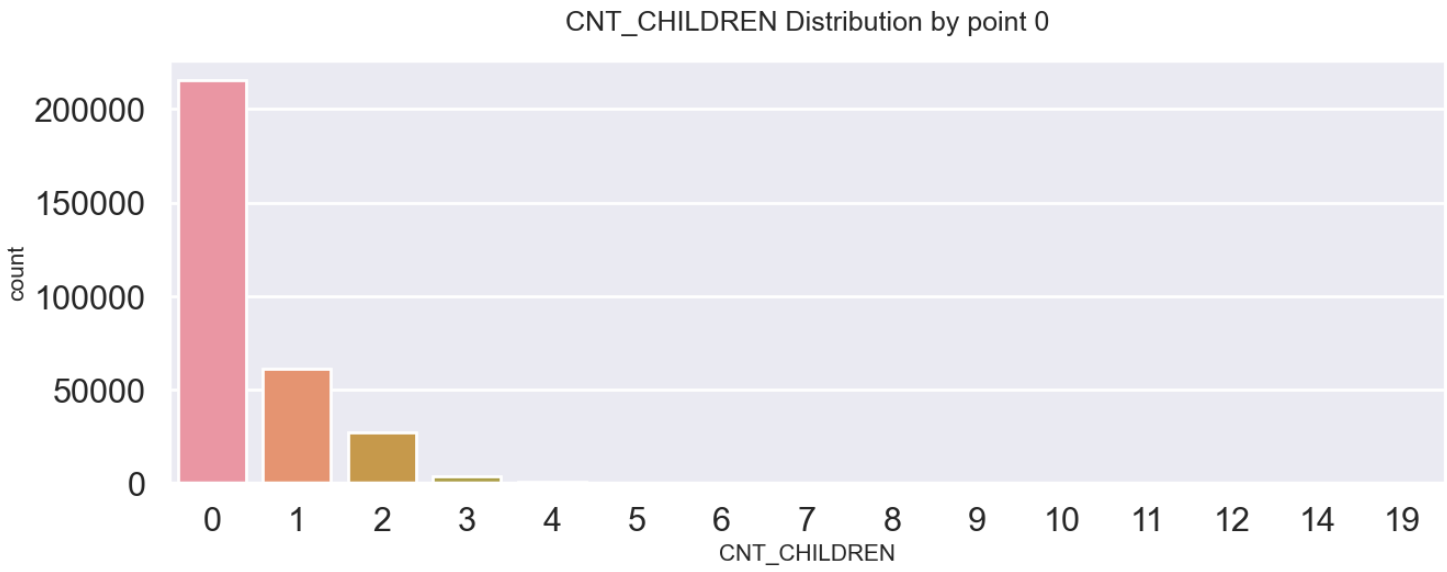
Points to be concluded from the above graph for Point=1 (Defaulters).

1. High number of credit for income type working , commercial associate , pensioner and state servant . Same as of target0

2. Low number of credit for income type unemployed and maternity leave

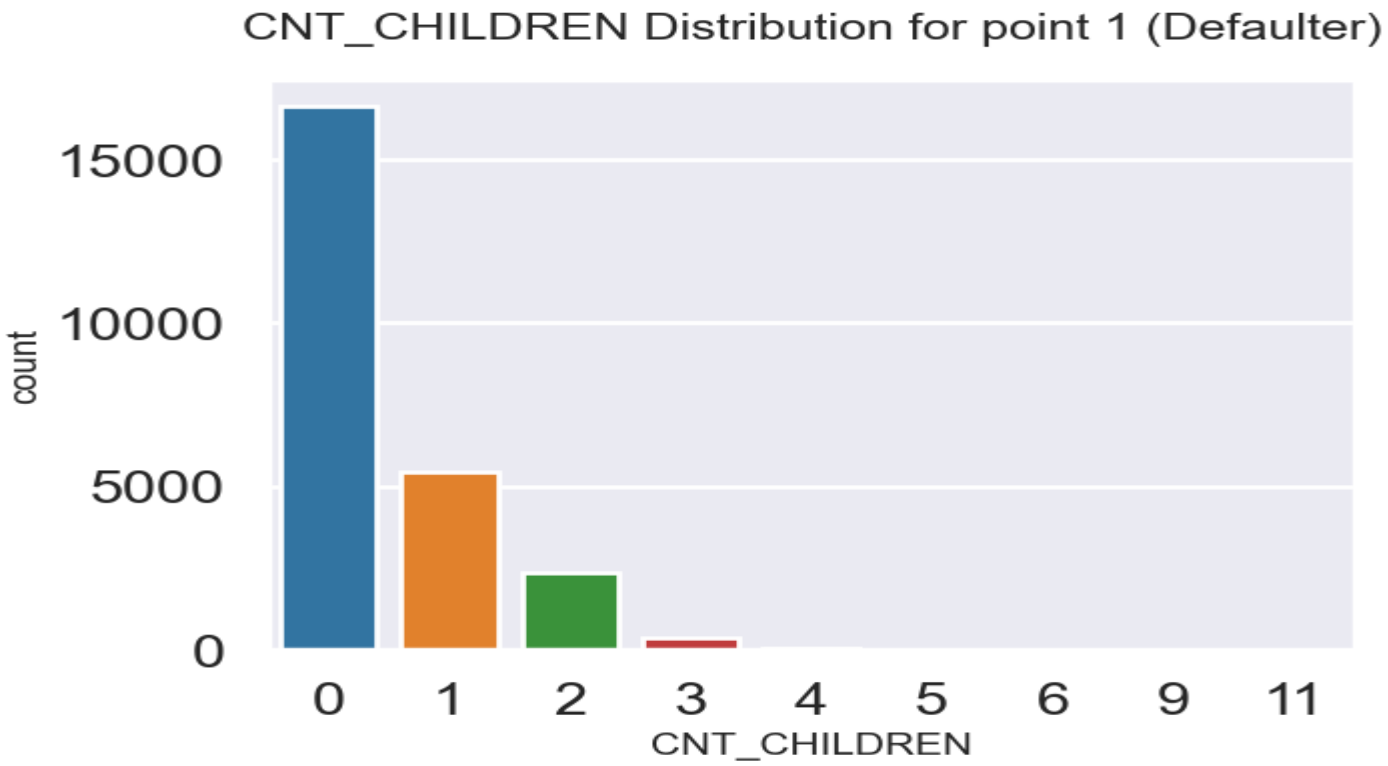


COUNT FOR CHILDREN FROM POINT 0 AND POINT 1



Points to be concluded from the beside graph:

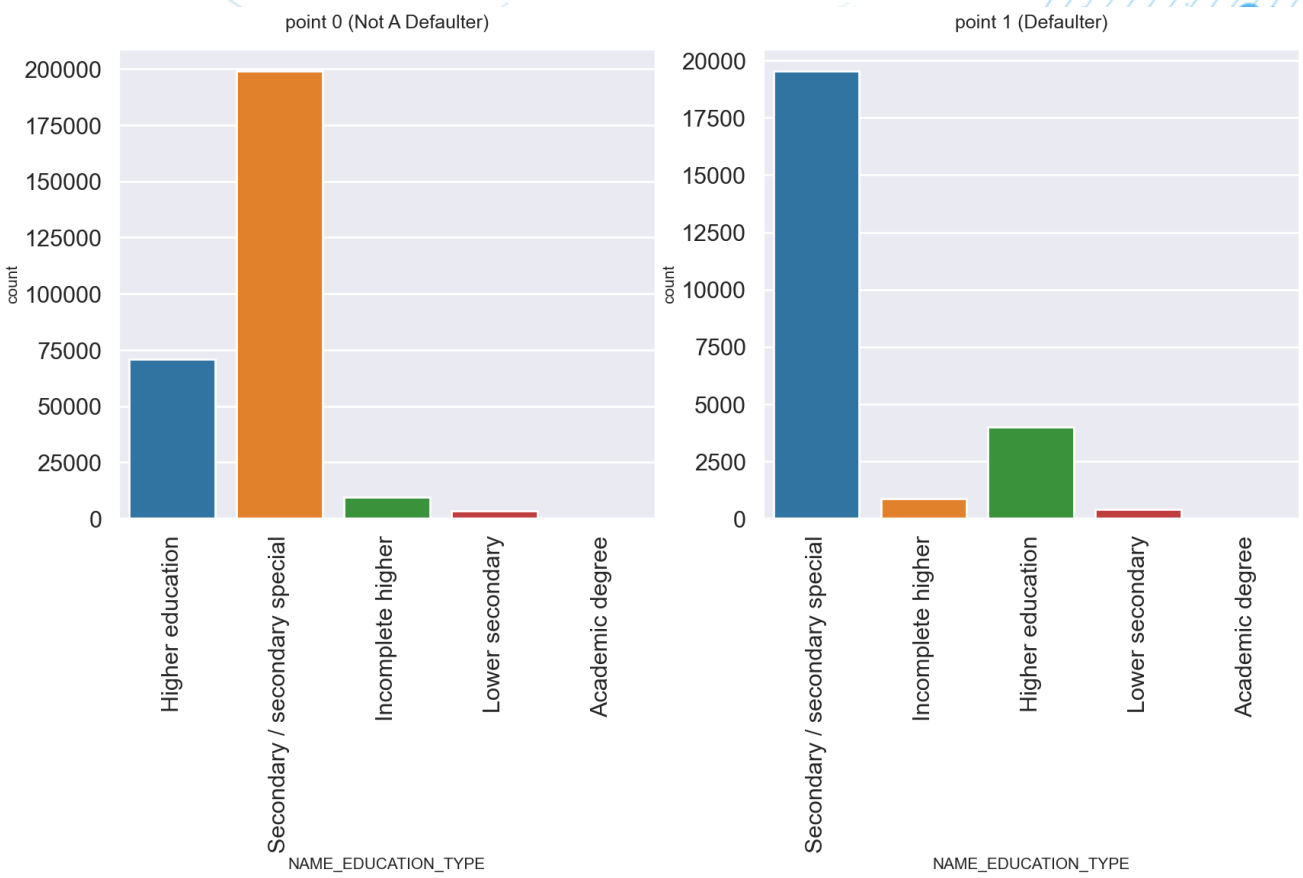
1. We can see that low child count maximizes that chances of both being a defaulter and also non defaulter. So we cannot conclude any specifics from this exploration.



NAME AND EDUCATION TYPE FROM POINT 0 AND POINT 1

Points to be concluded from the beside graph :

1. From this comparison we can see that people with secondary education has defaulted the most.

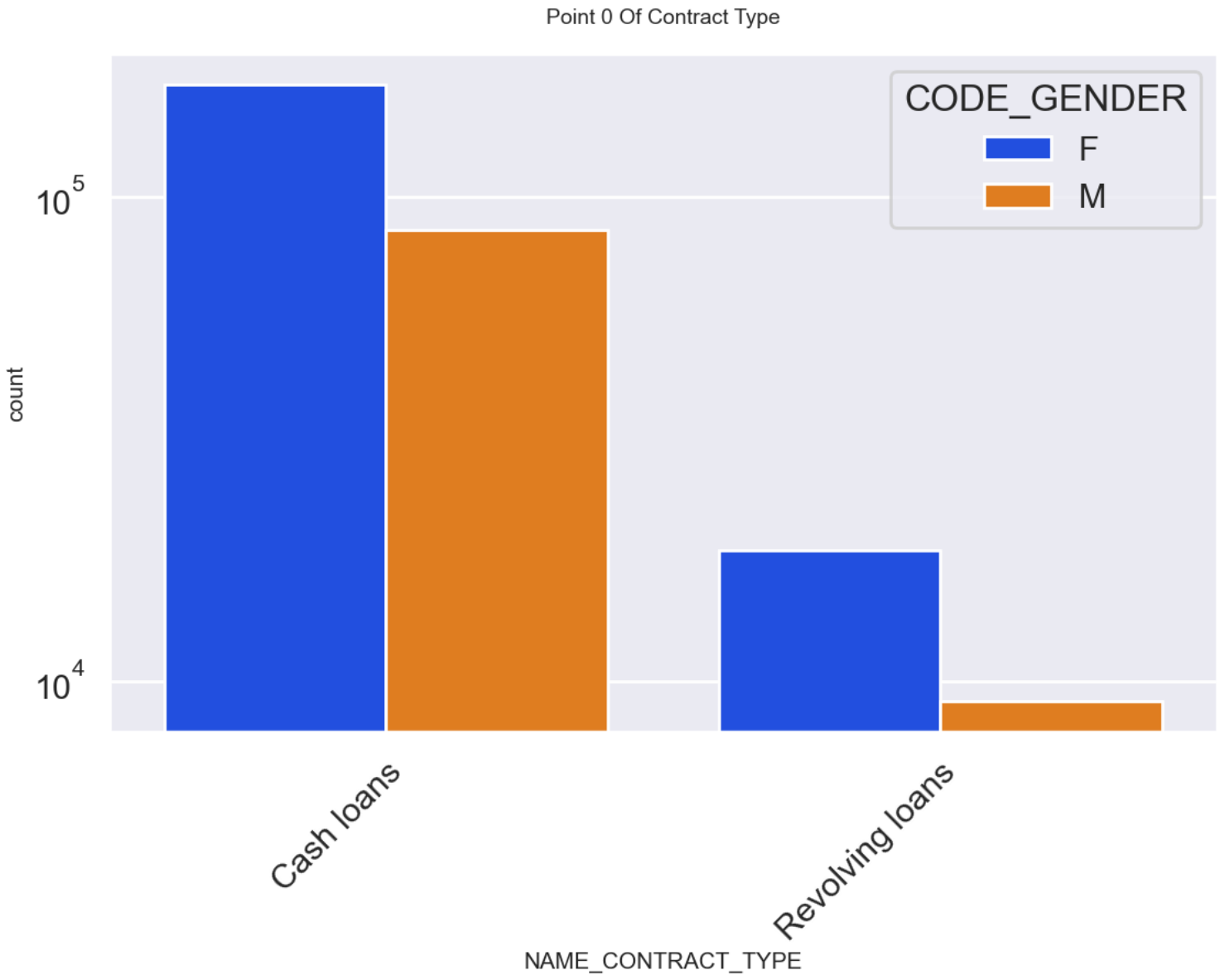


NAME OF THE CONTACT TYPE FOR POINT O

Points to be concluded from the beside graph for Point=0 (Non-Defaulters).

1. Cash Loan contracts have a higher number of credit than revolving loan contracts

2. Count of female is more

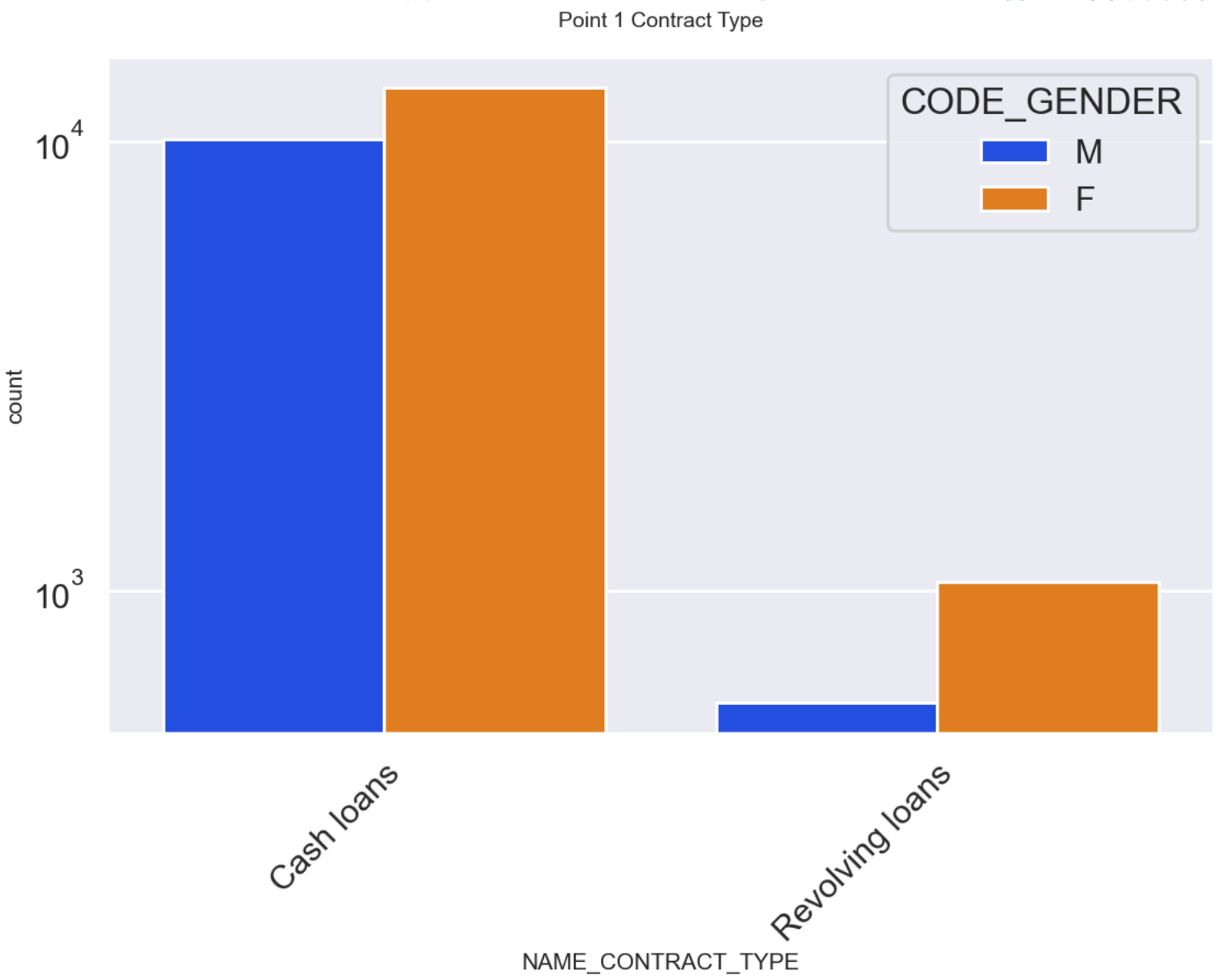


NAME OF THE CONTACT TYPE FOR POINT 1

Points to be concluded from the beside graph for Point = 1 (Defaulters).

1. Cash Loan contracts have a higher number of credit than revolving loan contracts

2. There is only female revolving loans.



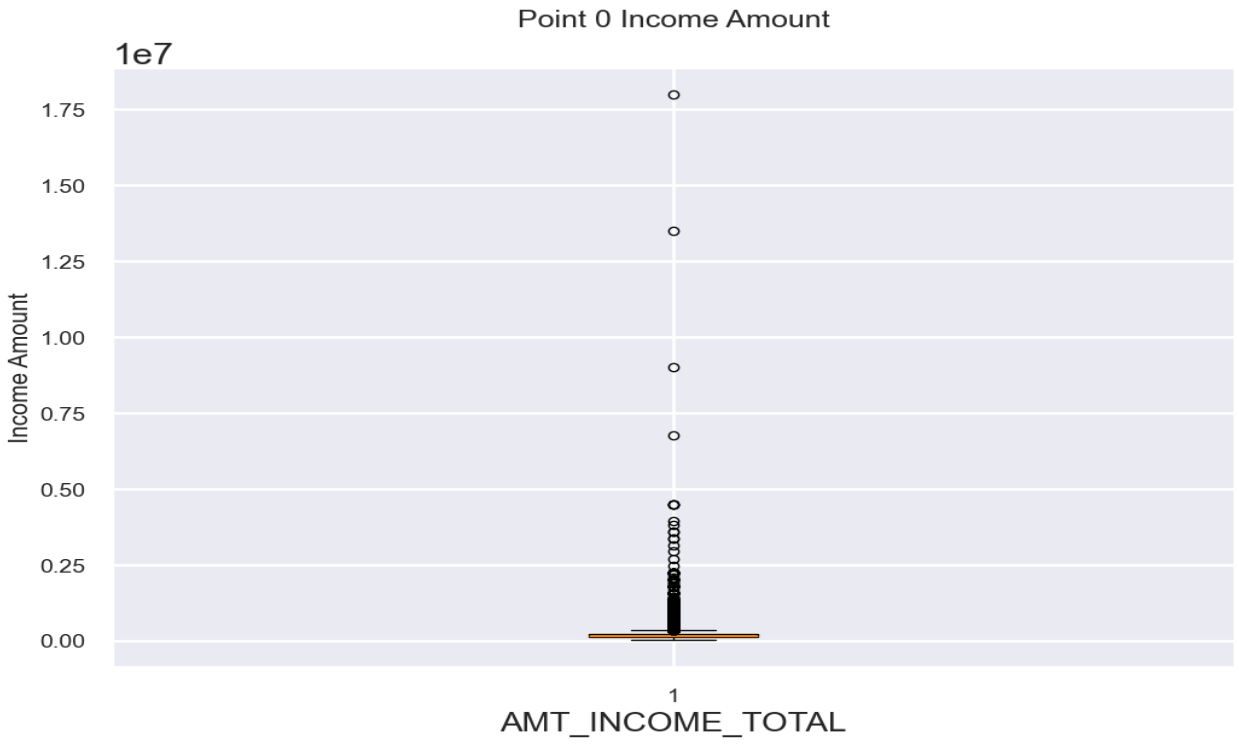
CONTINOUS UNIVARIATE ANALYSIS

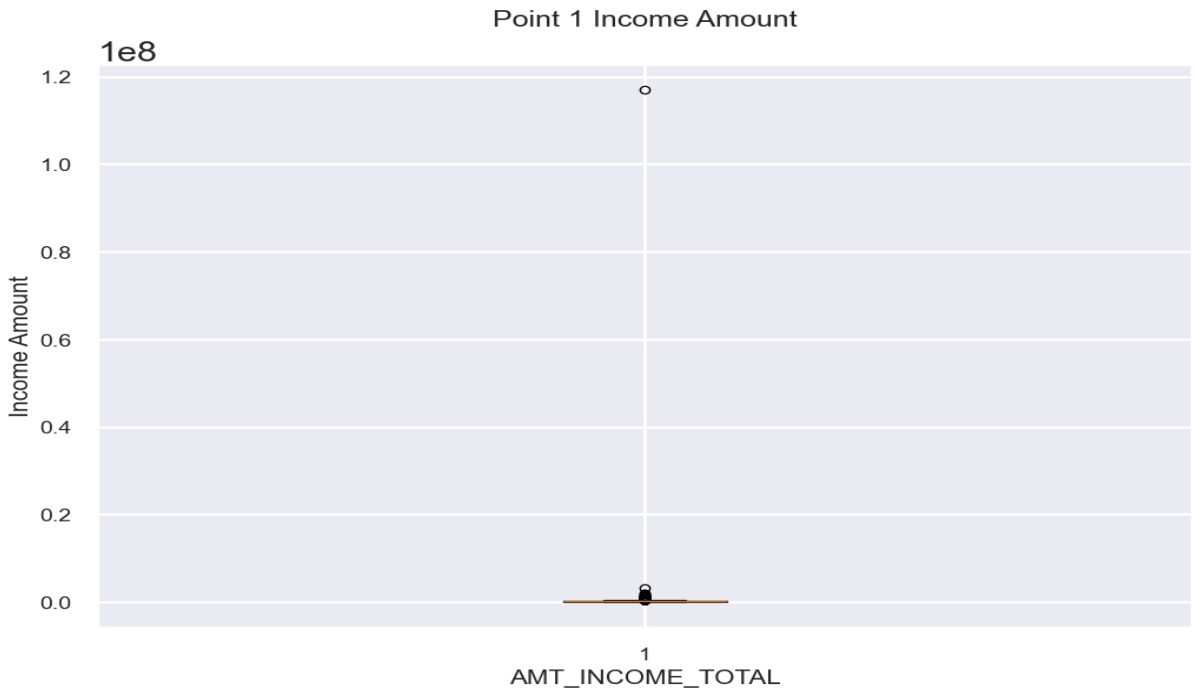
Points to be concluded from the beside 2 graph

1. Outliners are present in both

2. 3rd quartile is narrow for both Point 1 and Point 0

3. Most of the clients have income in the 1st quartile





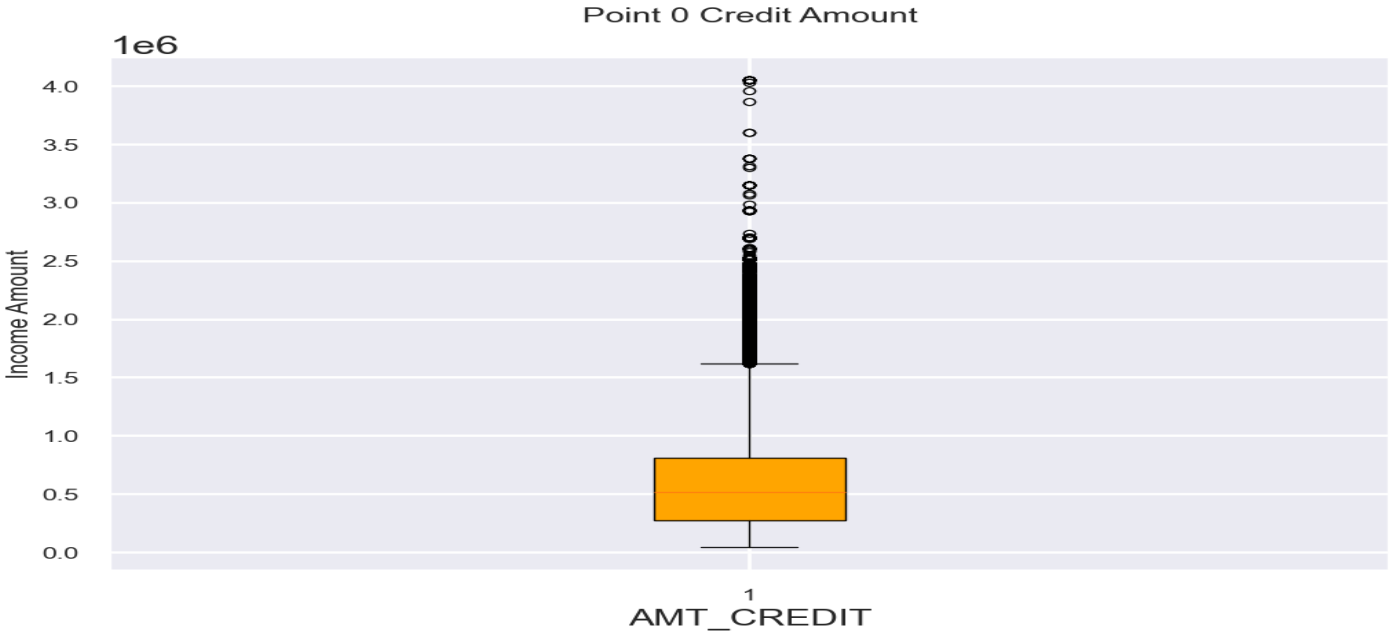
DISTRIBUTION OF CREDIT AMOUNT FOR POINT 1 & 0

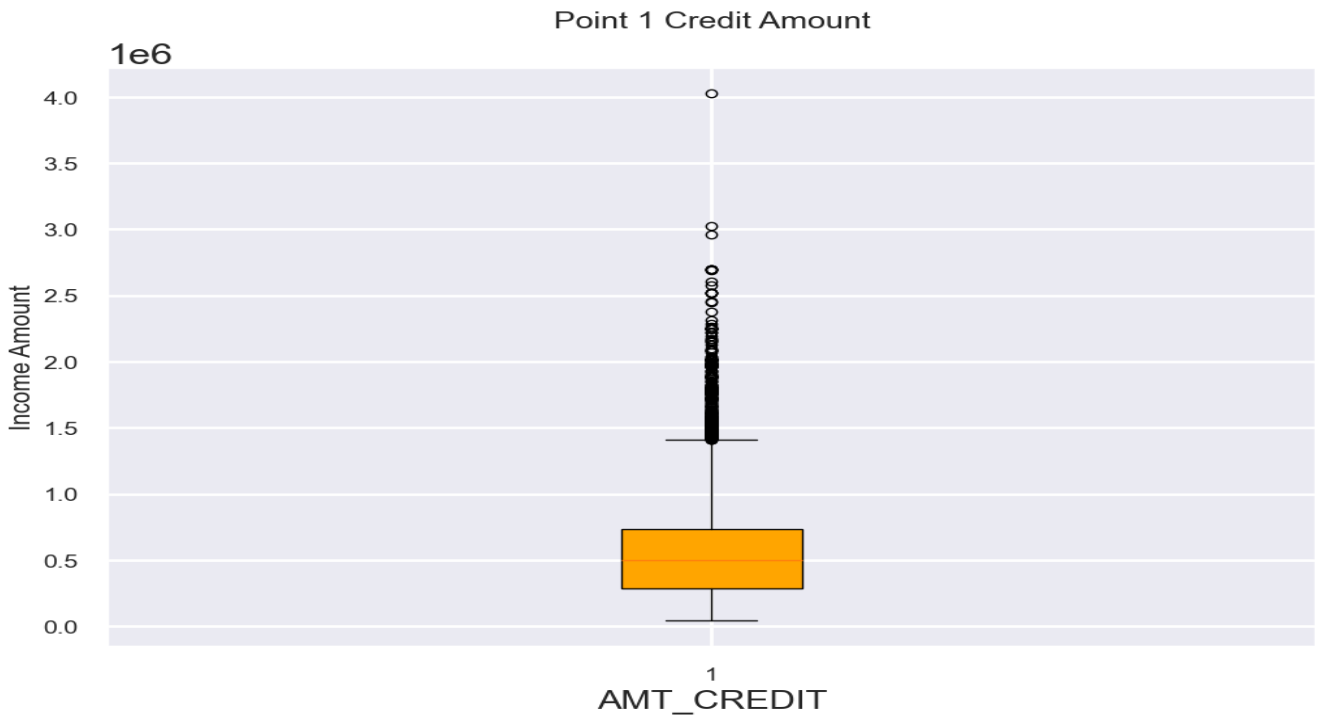
Points to be concluded from the beside 2 graph

1. Outliners are present in both

2. 3rd quartile is narrow for both Point 1 and Point 0

3. Most of the clients have credit amount in the 1st quartile

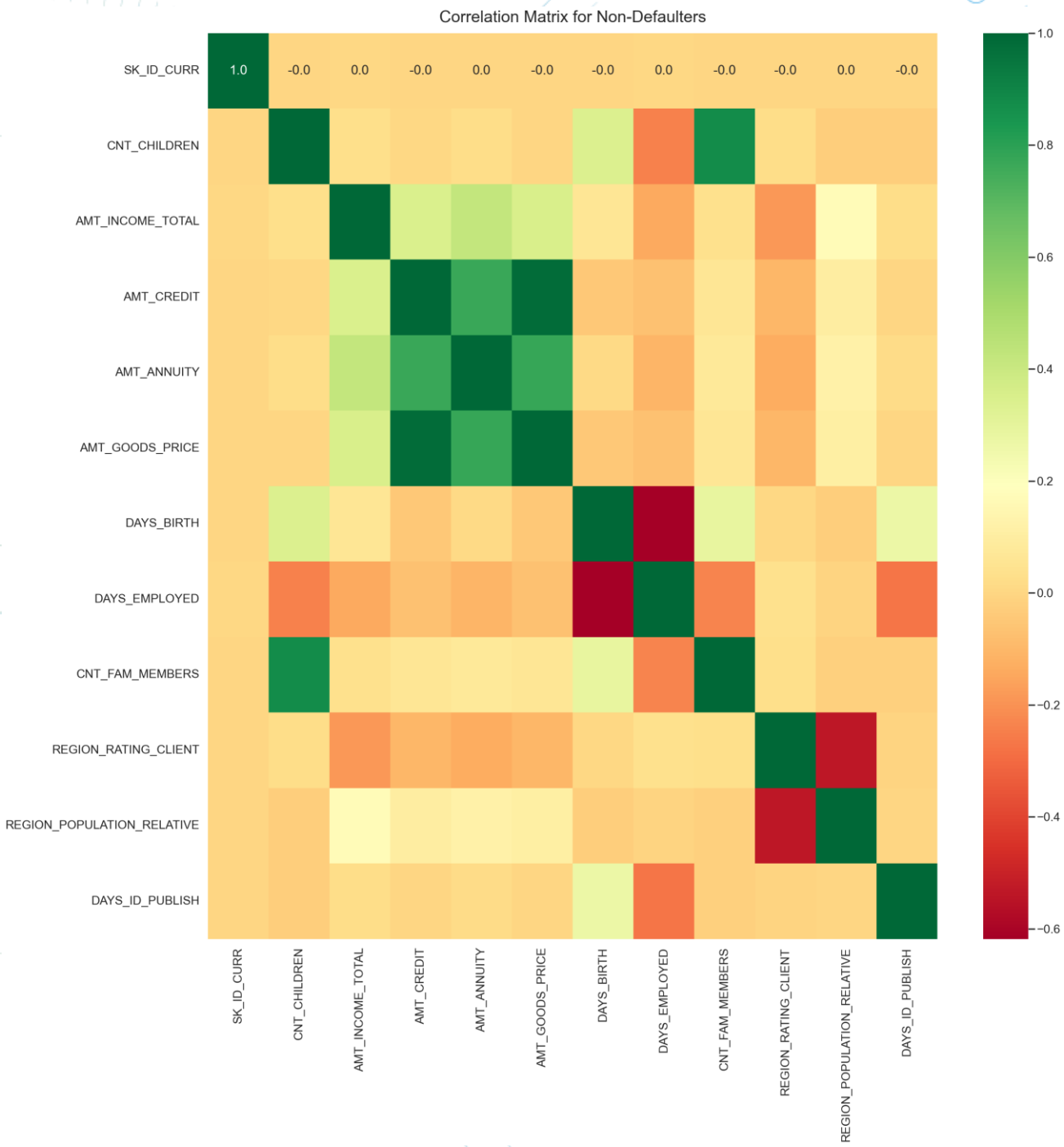




BIVARIATE ANALYSIS

These columns have high correlation values for Point 0.

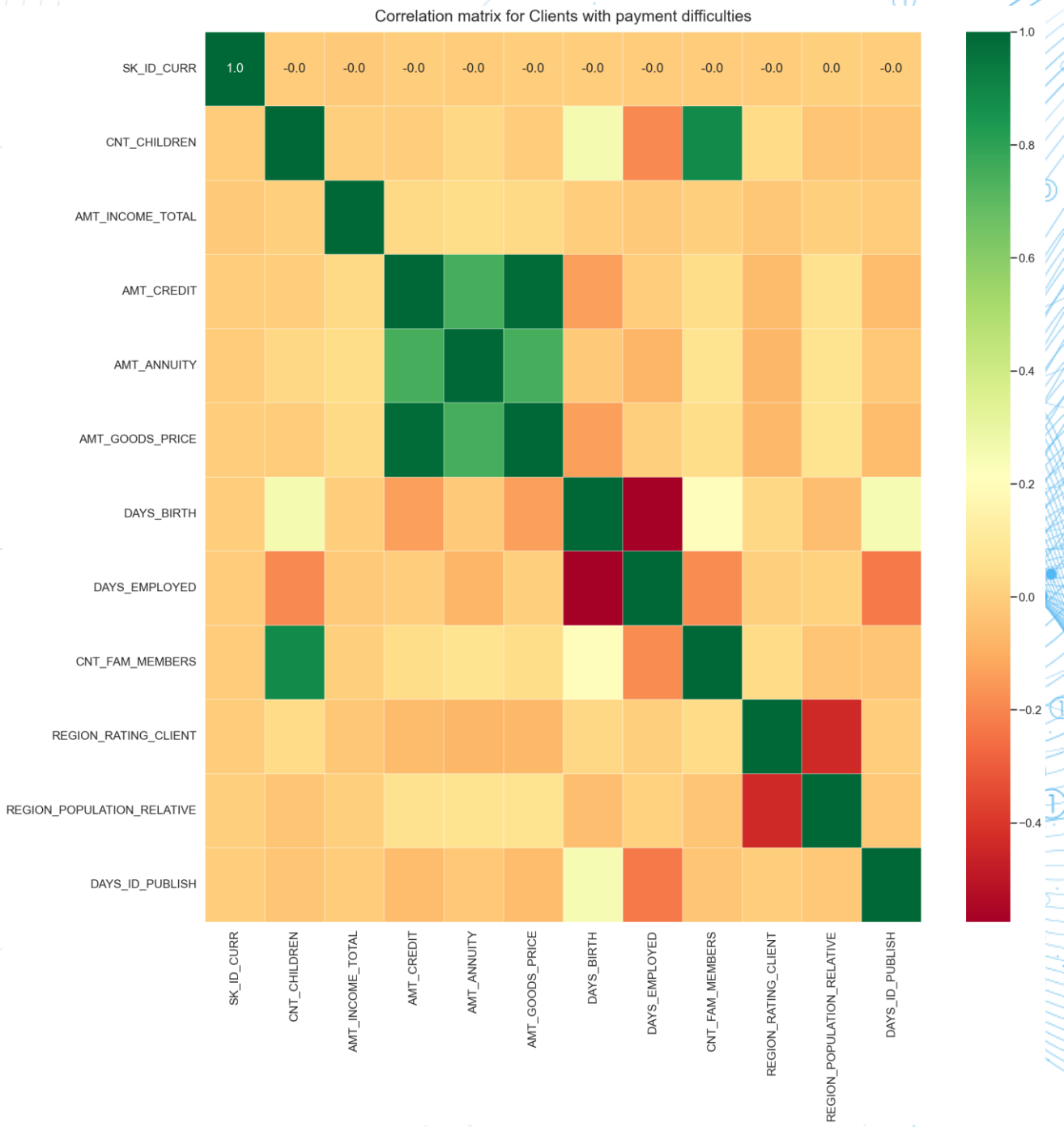
AMT_GOODS_PRICE and AMT_CREDIT
AMT_ANNUITY and AMT_CREDIT
AMT_ANNUITY and AMT_GOODS_PRICE
CNT_FAM_MEMBERS and CNT_CHILDREN
AMT_ANNUITY and AMT_INCOME_TOTAL
AMT_INCOME_TOTAL and AMT_GOODS_PRICE



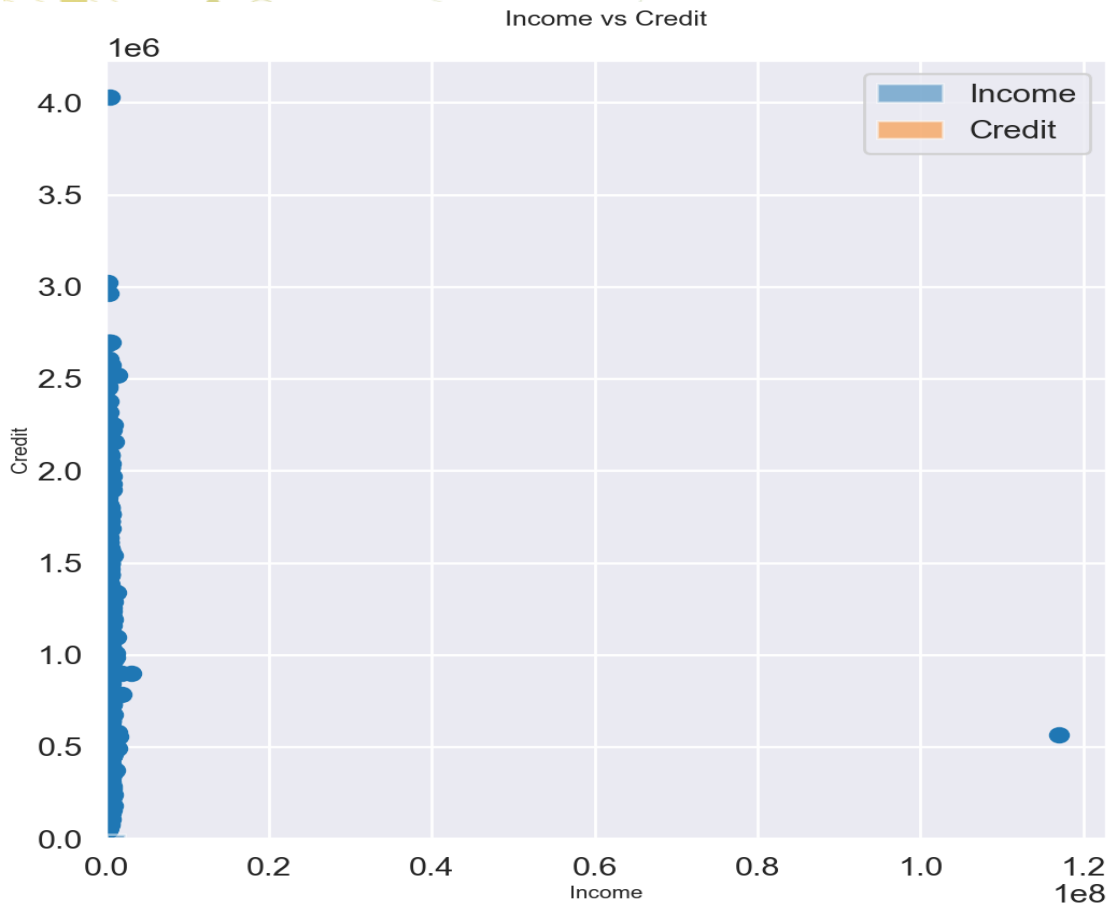
BIVARIATE ANALYSIS

Both for Point 0 and Point 1 these columns have high correlation values.

AMT_GOODS_PRICE and AMT_CREDIT
AMT_ANNUITY and AMT_CREDIT
AMT_ANNUITY and AMT_GOODS_PRICE
CNT_FAM_MEMBER and CNT_CHILDREN
AMT_ANNUITY and AMT_INCOME_TOTAL
AMT_INCOME_TOTAL and AMT_GOODS_PRICE

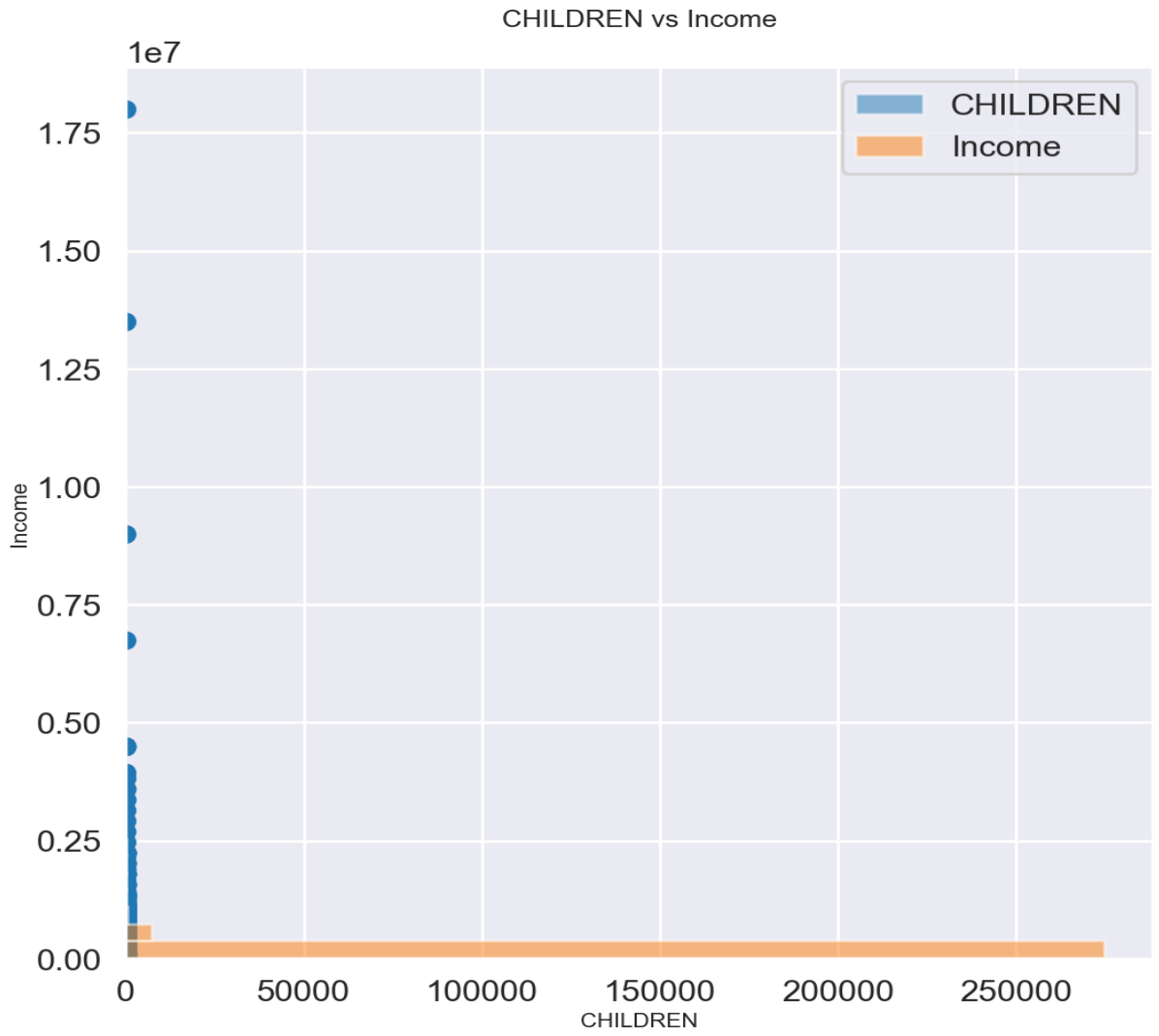


PLOTING INCOME VS CREDIT FOR POINT 1

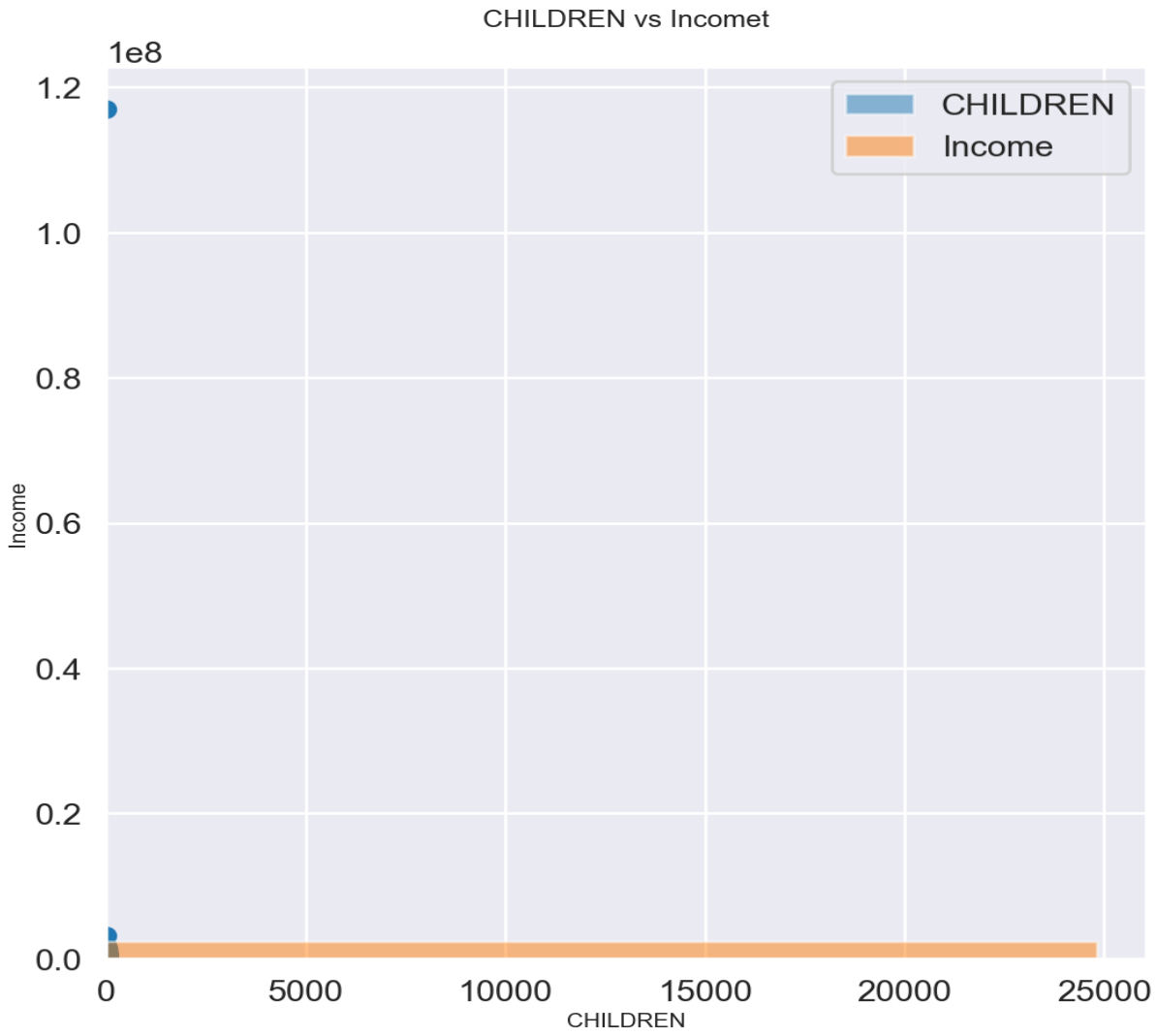


PLOTING INCOME VS CREDIT FOR POINT 0

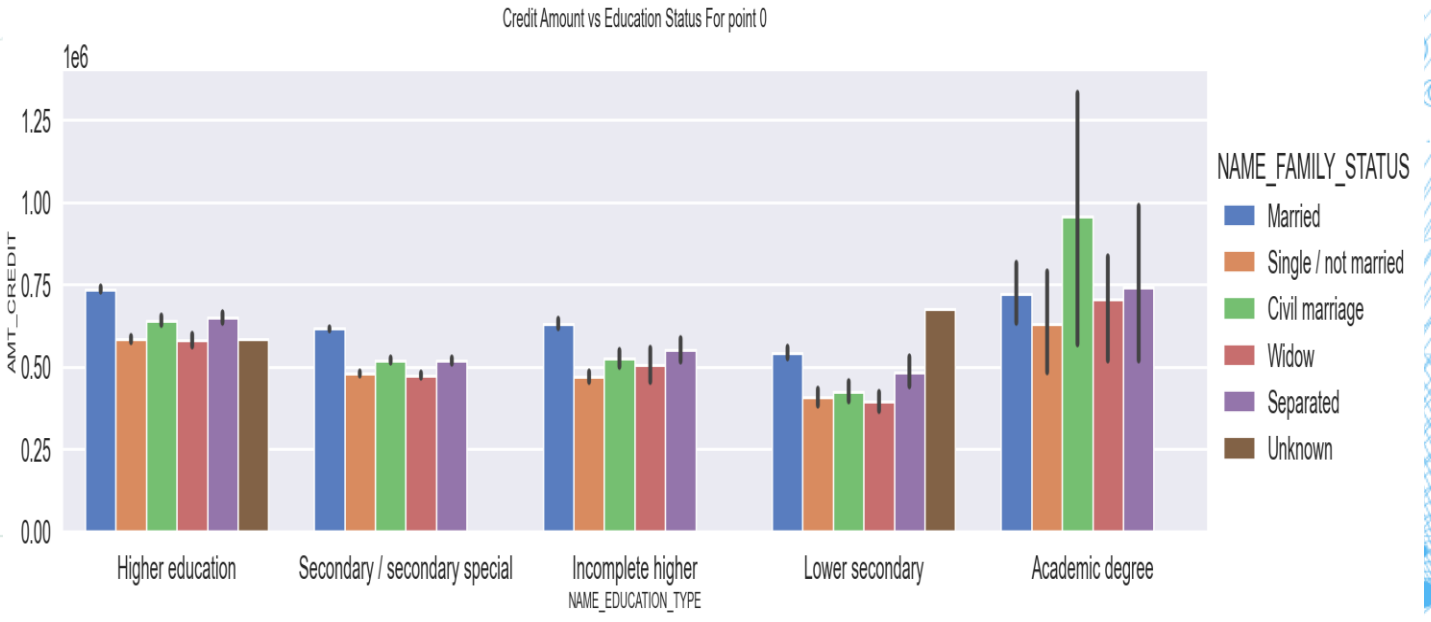
PLOTING CHILDREN VS INCOME FOR POINT 0



PLOTING CHILDREN VS INCOME FOR POINT 1



ANALYSIS OF CREDIT AMOUNT WITH RESPECT TO EDUCATION STATUS



Points to be concluded from the above graph Point = 0 (Non-Defaulters).

1. Customers holding academic degree have greater credit amount, Civil marriage segment being the highest among them.

2. Lower educated customers tends to have lower credit amount, Widows being the lowest among them

3. Married customers in almost all education segment except lower secondary and academic degrees have a higher credit amount.

ANALYSIS OF CREDIT AMOUNT WITH RESPECT TO EDUCATION STATUS

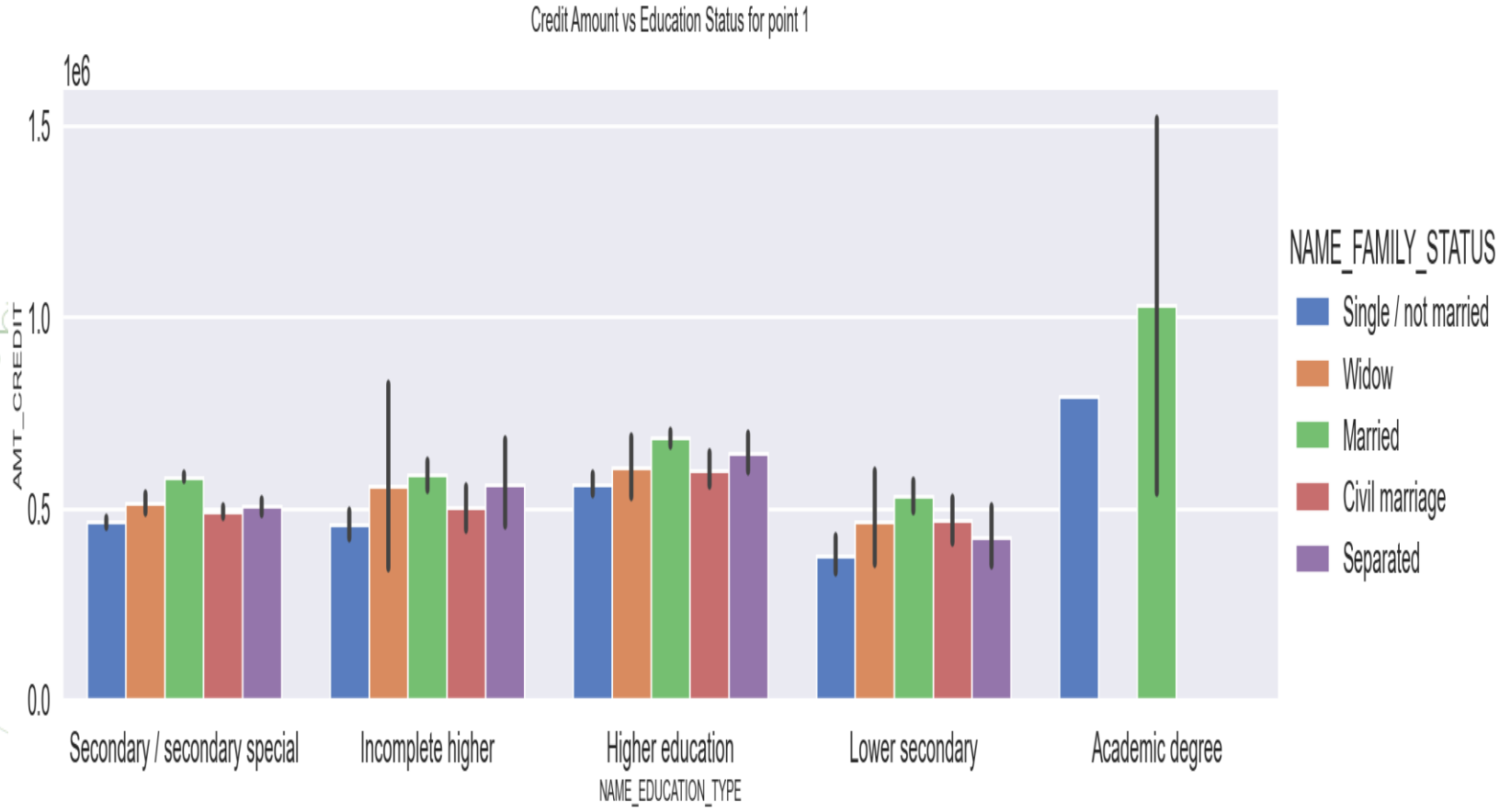
Points to be concluded from the above graph for Point = 1 (Defaulters).

1. Married Academic degree holding customers generally have a higher credit amount and so their defaulting rate is also high

2. Across all education segment married customer tends to have higher credit amount

3. Customers holding lower education tends to have a lower credit amount.

4. Single and Married are the only 2 family types present in academic degree .



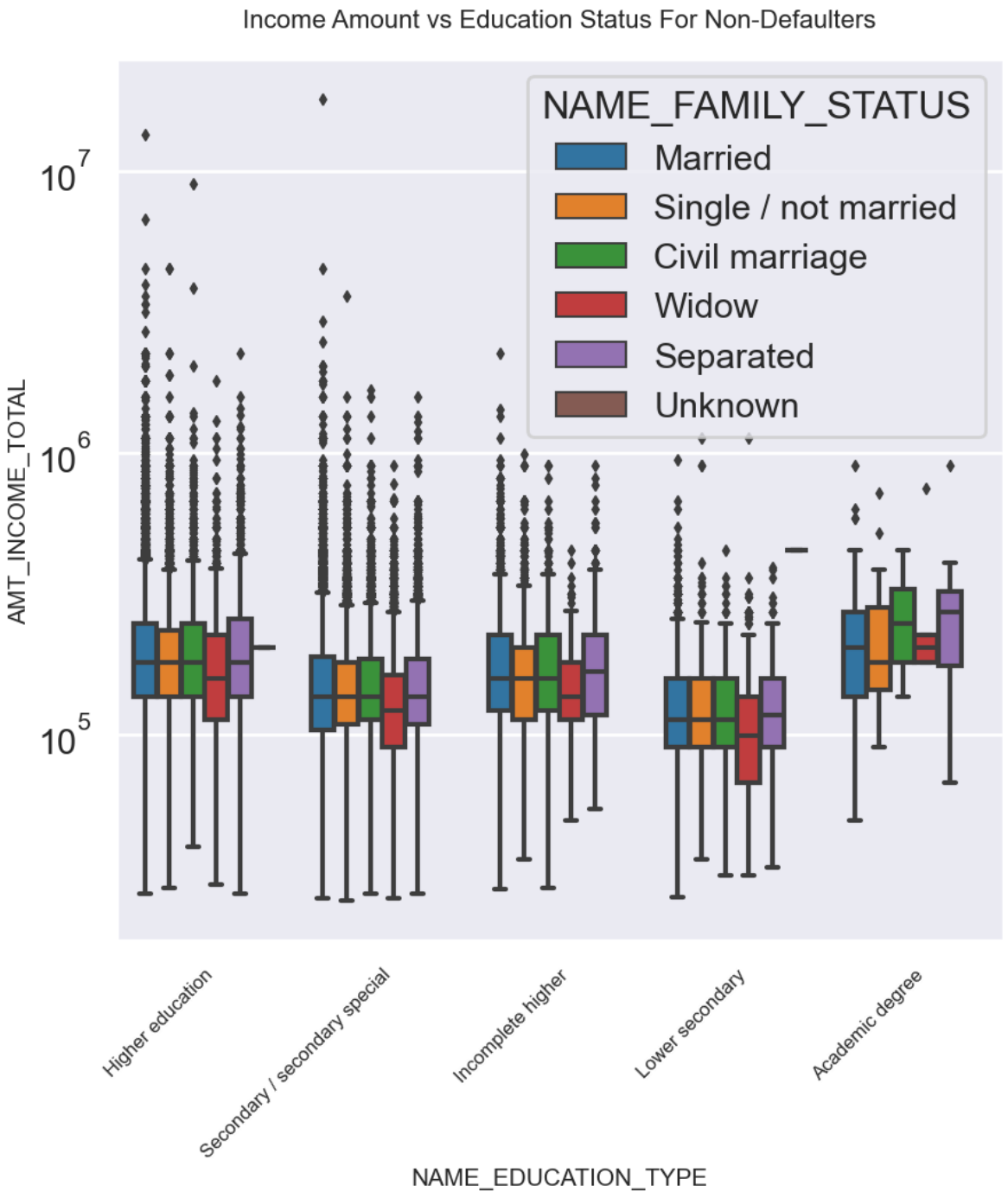
Analysis of Income amount with respect to Education Status

Points to be concluded from the above graph for point = 0 (Non-Defaulters).

1. For Education type 'Higher education' the income amount mean is mostly equal with family status. It does contain many outliers.

2. Less outlier are having for Academic degree but they are having the income amount is little higher that Higher education.

3. Lower secondary of civil marriage family status are have less income amount than others.



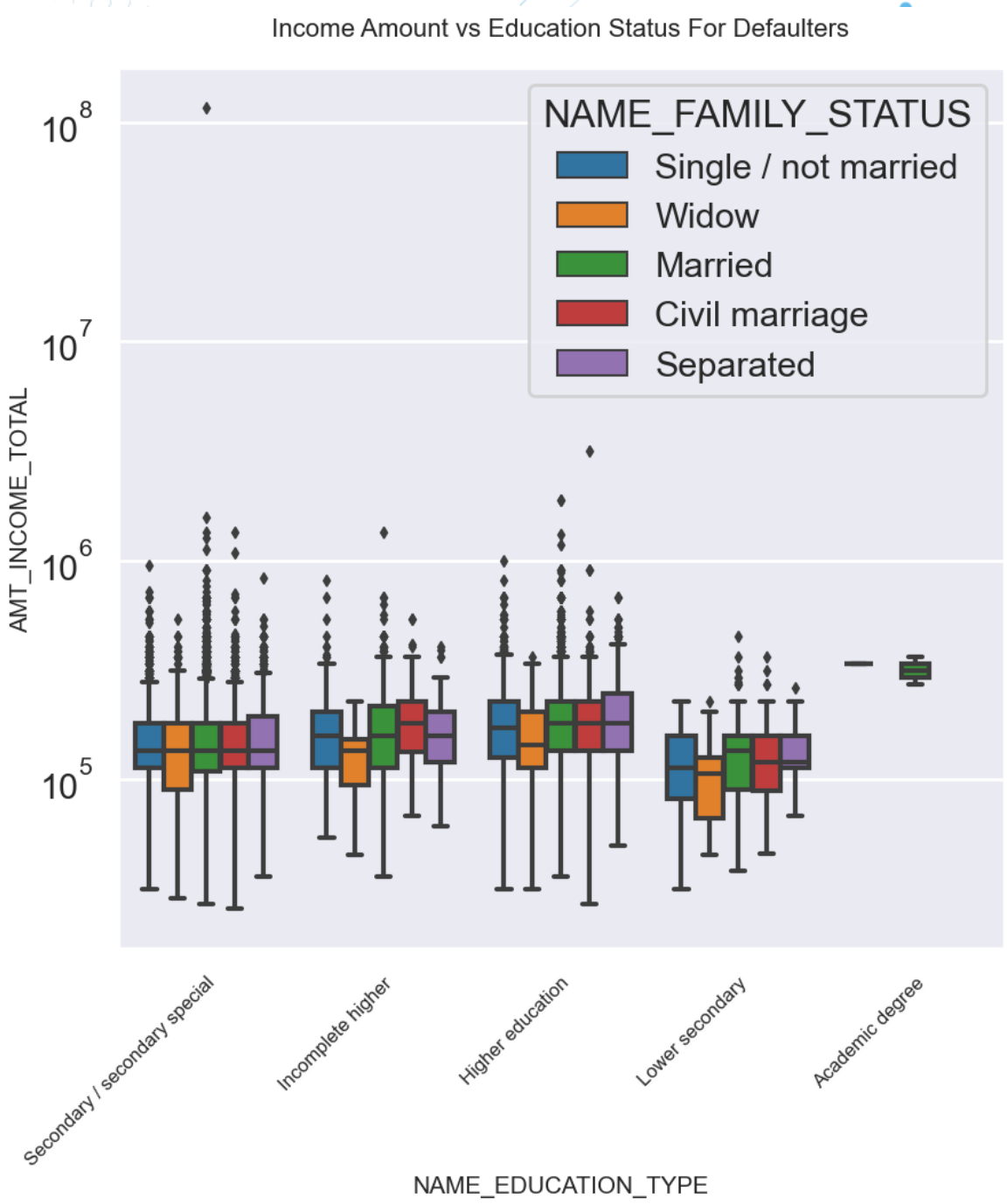
Analysis of Income amount with respect to Education Status

Points to be concluded from the above graph for Point = 1 (Defaulters).

1. Have some similarity with Target0, From above boxplot for Education type 'Higher education' the income amount is mostly equal with family status.

2. No outlier for Academic degree but there income amount is little higher than that Higher education.

3. Lower secondary are having less income amount than others.

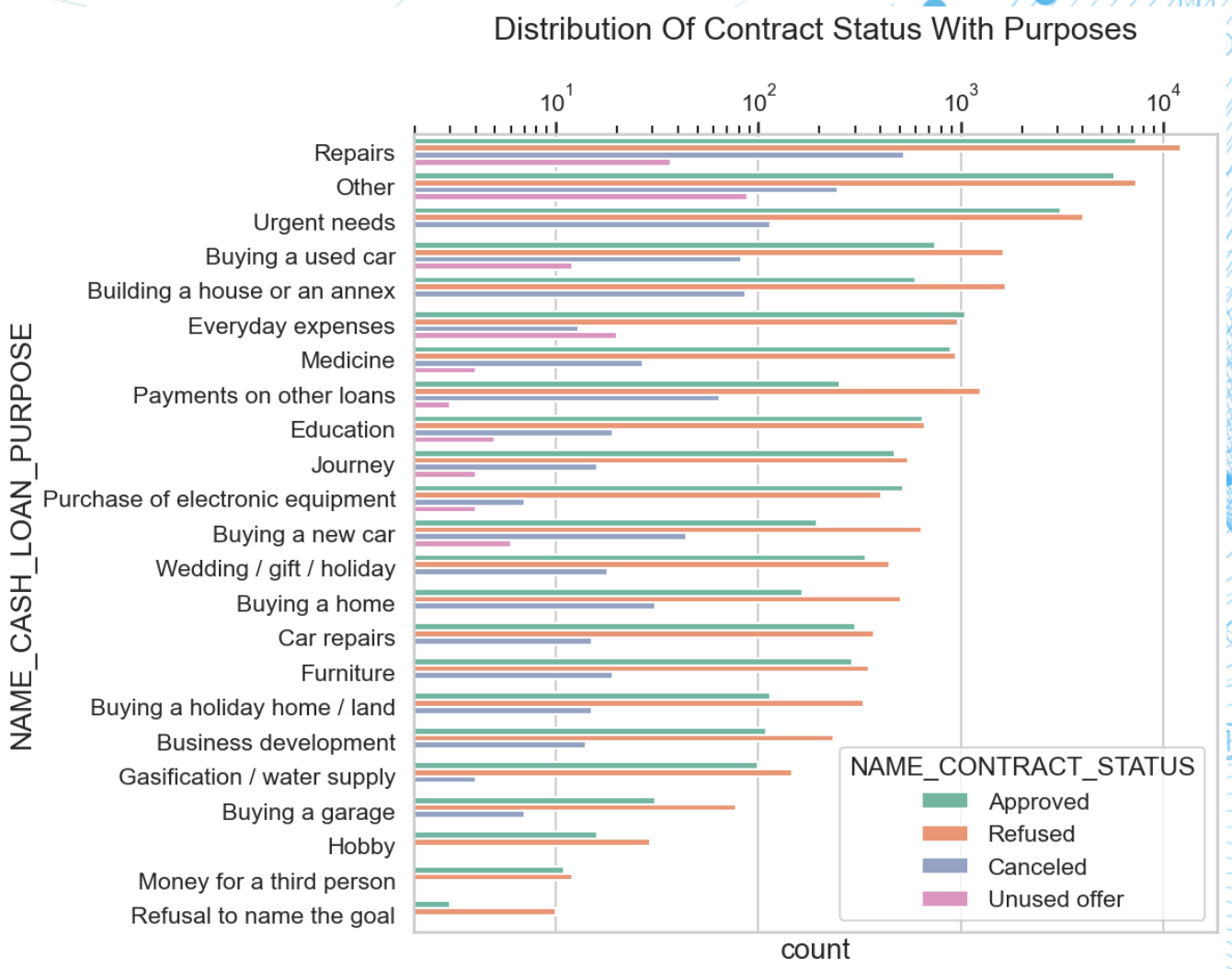


UNIVARIATE ANALYSIS

We can conclude the below points from the graph:

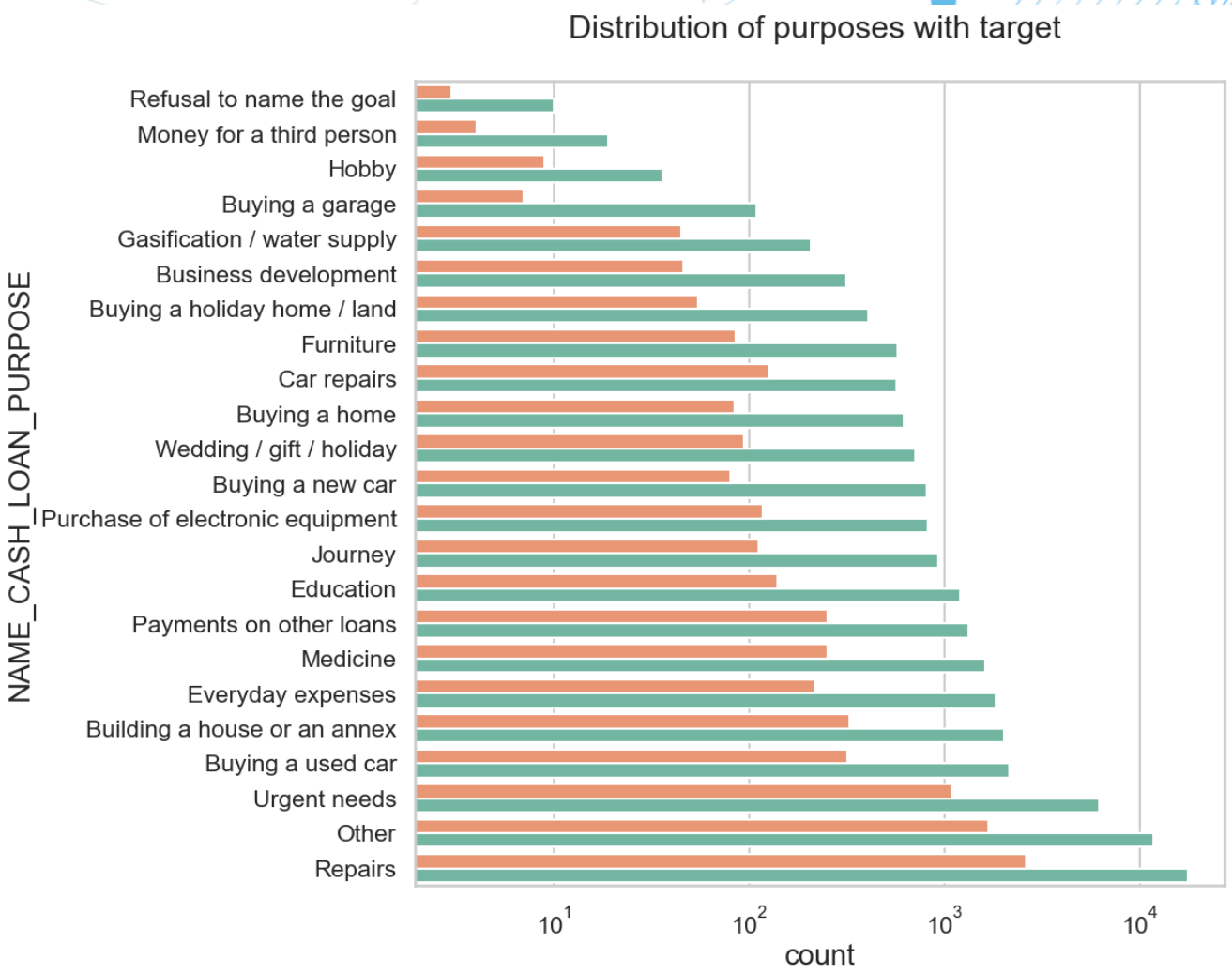
1.Most rejection of loans came from purpose 'Repairs'.

2.We have almost equal number of approves and rejection for Medicine , Every day expenses and education purposes.



UNIVARIATE ANALYSIS

We can conclude from above plot that Loan purposes with 'Repairs' are facing more difficulties in payment on time.



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